



Faculty of Engineering & Technology
Electrical & Computer Engineering Department
Computer Networks – ENCS3320
Project 2 Report
(Cisco Packet Tracer)

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Abstract:

This project focuses on designing, simulating, and analyzing a comprehensive computer network using Cisco Packet Tracer. The primary objectives include mastering subnetting techniques, implementing static and dynamic routing protocols (OSPF and BGP), configuring essential network services such as Web, Email, DNS, and DHCP, and securing wireless LANs. The project encompasses three autonomous systems (AS): Google, Faculty of Engineering and Technology, and Home-ISP, interconnected to simulate real-world network scenarios.

The project involves designing IP addressing schemes, configuring network devices, and ensuring seamless communication across all subnets and ASs. Advanced features like dynamic NAT with PAT and secure wireless configurations are implemented to enhance connectivity and security. The project emphasizes teamwork, requiring collaborative design, implementation, and documentation of network setups.

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Theory and Procedure :

Subnetting:

Subnetting is a technique that divides an IP network into smaller, more manageable networks known as subnets. This approach enhances routing efficiency, improves network security, and reduces broadcast traffic.

Subnetting focuses on the IP address, which is made up of two parts, namely a network part and a host part. The network part identifies the subnet to which the IP address belongs, while the host part identifies the specific device to which the IP address is assigned.

Subnetting allows you to use bits from the host part to extend the network part. In essence, it creates smaller subnetworks from a larger network.

Let's say we have the IP address 192.168.1.0 with a subnet mask of 255.255.255.0. This subnet mask indicates that the first 24 bits are the network part, and the last 8 bits are for hosts.

Therefore, we can have 256 IP addresses (0 to 255). Of those, two addresses are reserved for the network address and the broadcast address, which leaves us with 254 usable IP addresses.

Using the subnetting approach, we can split this network into smaller subnets. By changing the subnet mask to `255.255.255.128`, we've effectively borrowed one bit from the host part to make 2 subnets, giving us two subnets: `192.168.1.0/25` and `192.168.1.128/25`. [1]

Static and Dynamic Routing:

Static routing and dynamic routing are two methods used to determine how to send a packet toward its destination.

Static routes are configured in advance of any network communication. Dynamic routing, on the other hand, requires routers to exchange information with other routers to learn about paths through the network. Static and dynamic routing are used where appropriate, and some networks use both.[2]

Static routing vs. dynamic routing		
	Static routing	Dynamic routing
PATH SELECTION	One pre-configured route to destination	Multiple available routes to destination
ROUTE UPDATES	Engineers must reconfigure to make route changes	Algorithms automatically update with preferred route changes
ROUTING TABLES	Smaller routing table with only one entry for each destination	Routers send out entire routing tables to identify route availability
PROTOCOLS AND ALGORITHMS	Does not use protocols or algorithms for pre-configured route	Distance vector algorithms (RIP, IGRP) and link state algorithms (OSPF, IS-IS) adjust routes
COMPUTATION AND BANDWIDTH	Requires less computation time and bandwidth	Requires more computation and bandwidth
SECURITY	Better security	Less security
USE CASES	Used in smaller networks with fewer routers and unchanging network architecture	Used in larger networks and in networks that change frequently

Figure 1: Static Routing vs. Dynamic Routing

1. Static Routing:

Static routing, or nonadaptive routing, involves manually configuring routes in a router. This method is ideal for networks where routes are predictable and unlikely to change.

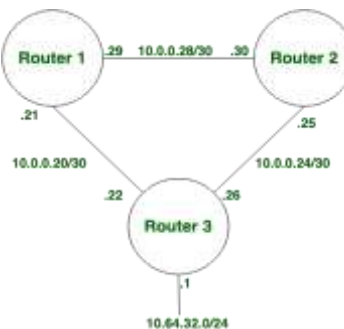


Figure 2: Static Routing

2. Dynamic Routing:

Dynamic routing, or adaptive routing, uses algorithms to automatically calculate and update the best paths for network traffic. It is suitable for larger or dynamic networks.

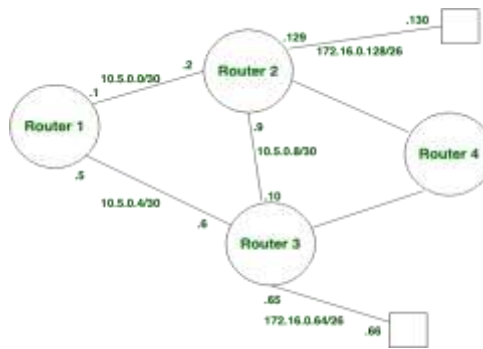


Figure 3:Dynamic Routing

Routing Protocols in Dynamic Routing:

a. Distance Vector Protocols:

Examples:

Routing Information Protocol (RIP), Interior Gateway Routing Protocol (IGRP).

Each router exchanges its entire routing table with directly connected routers periodically. Routing decisions are based on the hop count.[2]

b. Link State Protocols:

Examples:

Open Shortest Path First (OSPF):

Open Shortest Path First (OSPF) is a link-state routing protocol that is used to find the best path between the source and the destination router using its own Shortest Path First). OSPF is developed by Internet Engineering Task Force (IETF) as one of the Interior Gateway Protocol (IGP), i.e, the protocol which aims at moving the packet within a large autonomous system or routing domain. It is a network layer protocol which works on protocol number 89 and uses AD value 110. OSPF uses multicast address 224.0.0.5 for normal communication and 224.0.0.6 for update to designated router(DR)/Backup Designated Router (BDR). [3]

OSPF is a link-state routing protocol featuring:

Multicast packet transmission, reducing load on the routers that do not run OSPF

Support for Classless Inter-Domain Routing (CIDR)

Load balancing among equal-cost routes

Packet authentication

With the preceding advantages, OSPF is widely accepted and used as an IGP.[5]

Multicast:

Multicast is one of the three IP transmission modes and allows IP packets sent by a source to be forwarded to a specific group of receivers. There is at most one copy of the same multicast data flow on each link. Compared with traditional unicast and broadcast, multicast effectively conserves network bandwidth and reduces network load. Therefore, multicast is widely used in network services, such as in IPTV, real-time data transmission, and multimedia conferencing.[6]

In this project, the OSPF protocol was utilized as one of the primary dynamic routing protocols to connect subnets within autonomous systems (AS). OSPF plays a crucial role in ensuring efficient routing between different networks by creating a complete map of the network and determining the optimal path using the Shortest Path First (SPF) algorithm.

OSPF was configured within each autonomous system, such as the Google Network (AS-300) and the Faculty of Engineering and Technology Network (AS-100), to route packets between internal subnets.

Area 0 (Backbone Area) was used to connect all subnets within each autonomous system.

OSPF was integrated with the BGP protocol to ensure packet routing between different autonomous systems (Inter-AS Routing), thereby guaranteeing complete connectivity among the three networks (Google, Home-ISP, and Faculty of Engineering).

Border Gateway Protocol (BGP):

The protocol can connect any internetwork of the autonomous system using an arbitrary topology. The only requirement is that each AS have at least one router that can run BGP and that is the router connected to at least one other AS's BGP router. BGP's main function is to exchange network reachability information with other BGP systems. Border Gateway Protocol constructs an autonomous systems graph based on the information exchanged between BGP routers.

Functionality of Border Gateway Protocol (BGP):

BGP peers perform 3 functions, which are given below.

The first function consists of initial peer acquisition and authentication. both the peers established a TCP connection and performed message exchange that guarantees both sides have agreed to communicate.

The second function mainly focuses on sending negative or positive reach-ability information.

The third function verifies that the peers and the network connection between them are functioning correctly.

Types of Border Gateway Protocol

External BGP: It is used to interchange routing information between the routers in different autonomous systems, it is also known as eBGP(External Border Gateway Protocol). The below image shows how eBGP interchange routing information.

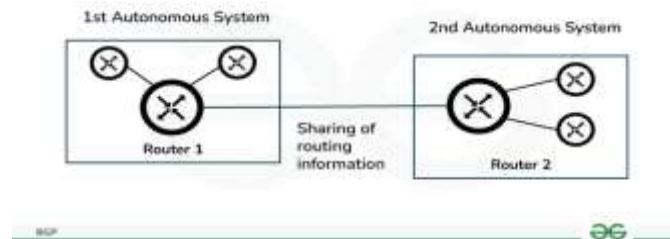


Figure 4:eBGP

Internal BGP: It is used to interchange routing information between the routers in the same autonomous system, it is also known as iBGP (Internal Border Gateway Protocol). Internal routers also ensure consistency among routers for sharing routing information. The below image shows how iBGP interchange routing information.[7]

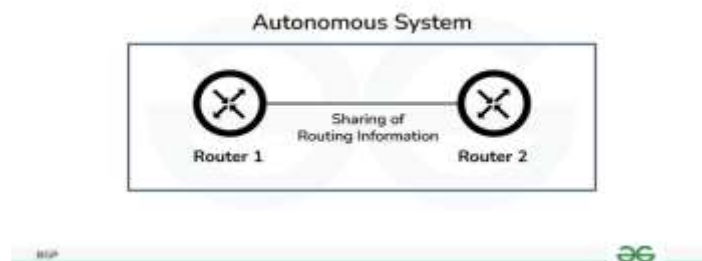


Figure 5:iBGP

Network Address Translation (NAT):

Network Address Translation (NAT) is a process in which one or more local IP addresses are translated into one or more Global IP addresses and vice versa to provide Internet access to the local hosts. It also does the translation of port numbers, i.e., masks the port number of the host with another port number in the packet that will be routed to the destination. It then makes the corresponding entries of IP address and port number in the NAT table.

Generally, the border router is configured for NAT i.e. the router which has one interface in the local (inside) network and one interface in the global (outside) network. When a packet traverse outside the local (inside) network, then NAT converts that local (private) IP address to a global (public) IP address. When a packet enters the local network, the global (public) IP address is converted to a local (private) IP address.

If NAT runs out of addresses, i.e., no address is left in the pool configured then the packets will be dropped and an Internet Control Message Protocol (ICMP) host unreachable packet to the destination is sent.

Inside refers to the addresses which must be translated. Outside refers to the addresses which are not in control of an organization. These are the network Addresses in which the translation of the addresses will be done.

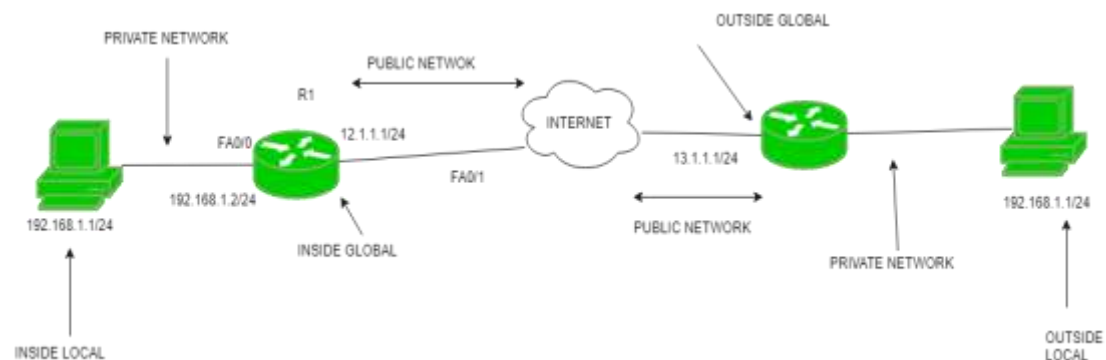


Figure 6:NAT

Inside local address – An IP address that is assigned to a host on the Inside (local) network. The address is probably not an IP address assigned by the service provider i.e., these are private IP addresses. This is the inside host seen from the inside network.

Inside global address – IP address that represents one or more inside local IP addresses to the outside world. This is the inside host as seen from the outside network.

Outside local address – This is the actual IP address of the destination host in the local network after translation.

Outside global address – This is the outside host as seen from the outside network. It is the IP address of the outside destination host before translation.[8]

Dynamic Host Configuration Protocol(DHCP):

The DHCP (Dynamic Host Configuration Protocol) is a network protocol used to automatically assign IP addresses and other configuration parameters to devices on a network. It simplifies the process of setting up and managing IP addresses by dynamically allocating them as devices connect to the network. This is especially useful in larger networks where manual IP address configuration would be time-consuming and error-prone.

Here's how DHCP works:

- **Request:** When a device (known as a DHCP client) connects to a network, it sends out a DHCP Discover message to locate a DHCP server on the network.
- **Offer:** DHCP servers on the network respond with a DHCP Offer message, providing the client with an available IP address, subnet mask, default gateway, DNS server addresses, and other configuration details.

- **Request:** The client then selects one of the offered configurations and sends a DHCP Request message to the server, indicating its choice.
- **Acknowledge:** The DHCP server responds with a DHCP Acknowledge message, confirming the allocation of the chosen configuration to the client.
- **Lease Renewal:** IP addresses assigned by DHCP are not permanent; they are leased for a specific period. Before the lease expires, the client can request a renewal from the DHCP server. If the server acknowledges the renewal, the lease is extended.
- **Release:** When a device disconnects from the network or no longer requires the assigned IP address, it sends a DHCP Release message to inform the server that the IP address can be returned to the pool of available addresses.[9]

How does DHCP work?

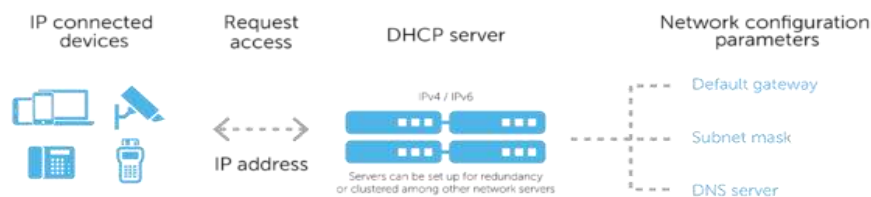


Figure 7:DHCP

The server will quickly and automatically assign an IP address and some related network configuration parameters. Once the device has accepted the assignment, it can communicate with both the internal network and the public internet.

Dynamic IP addressing with DHCP:

The assignment of IP addresses happens dynamically within a given address range. As a result, a device connected to the network doesn't have a forever address. The IP address can periodically change as its lease time expires unless the lease is successfully renewed.

For services that always need to be on, a static IP address is often a better option. Corporate enterprises commonly use static IP addresses for hardware like mail servers. Certainly, a DHCP server should have a static IP address.

However, there are drawbacks to dedicating a specific IP address to a device or service. A network administrator must manually assign, configure, and track the IP address. It's a time-consuming task. Oftentimes, it requires the admin to physically be with the device.

- 1- Meanwhile, dynamic IP addresses are usually the preferred choice because they:
- 2- Cost less to manage than static IP addresses;
- 3- May offer more privacy and security with a constantly changing IP address; and
- 4- Don't require manual administration when a device roams from one subnet to another.[4]

Web Server:

A web server is a computer system capable of delivering web content to end users over the internet via a web browser.

The end user processes a request via a web browser installed on a web server. The communication between a web server or browser and the end user takes place using Hypertext Transfer Protocol (HTTP). The primary role of a web server is to store, process, and deliver requested information or webpages to end users. It uses:

Physical Storage: All website data is stored on a physical web server to ensure its safety. When an end user enters the URL of your website or searches it using a keyword on a browser, a request is generated and sent to the web server to process the data.

Web Browser: The role of web browsers such as Firefox, Chrome, or Internet Explorer is to find the web server on which your website data is located. Once the browser finds your server, it reads the request and processes the information.

Web servers are primarily used to process and manage HTTP/HTTPS requests and responses from the client system.[10]

Domain Name System (DNS):

The Domain Name System (DNS) turns domain names into IP addresses, which browsers use to load internet pages. Every device connected to the internet has its own IP address, which is used by other devices to locate the device. DNS servers make it possible for people to input normal words into their browsers, such as Fortinet.com, without having to keep track of the IP address for every website.

A DNS server is a computer with a database containing the public IP addresses associated with the names of the websites an IP address brings a user to. DNS acts like a phonebook for the internet. Whenever people type domain names, like Fortinet.com or Yahoo.com, into the address bar of web browsers, the DNS finds the right IP address. The site's IP address is what directs the device to go to the correct place to access the site's data.

Once the DNS server finds the correct IP address, browsers take the address and use it to send data to content delivery network (CDN) edge servers or origin servers. Once this is done, the information on the website can be accessed by the user. The DNS server starts the process by finding the corresponding IP address for a website's uniform resource locator (URL).

In a usual DNS query, the URL typed in by the user has to go through four servers for the IP address to be provided. The four servers work with each other to get the correct IP address to the client, and they include:

DNS recursor: The DNS recursor, which is also referred to as a DNS resolver, receives the query from the DNS client. Then it communicates with other DNS servers to find the right IP address. After the resolver retrieves the request from the

client, the resolver acts like a client itself. As it does this, it makes queries that get sent to the other three DNS servers: root nameservers, top-level domain (TLD) nameservers, and authoritative nameservers.

Root nameservers: The root nameserver is designated for the internet's DNS root zone. Its job is to answer requests sent to it for records in the root zone. It answers

requests by sending back a list of the authoritative nameservers that go with the correct TLD.

TLD nameservers: A TLD nameserver keeps the IP address of the second-level domain contained within the TLD name. It then releases the website's IP address and sends the query to the domain's nameserver.

Authoritative nameservers: An authoritative nameserver is what gives you the real answer to your DNS query. There are two types of authoritative nameservers: a master server or primary nameserver and a slave server or secondary nameserver. The master server keeps the original copies of the zone records, while the slave server is an exact copy of the master server. It shares the DNS server load and acts as a backup if the master server fails. [11]

Email Server :

An email server, also called a mail server, is essentially a computer system that sends and receives emails. When you send an email, it goes through a series of servers to reach its final destination. While this process is lightning fast and efficient (thanks, technology!), there is a significant amount of complexity behind sending and receiving emails.

Email communication involves complex protocols and processes. Usually, the email server is a computer or machine that has a complete system with different applications or services. Based on the type of action they perform, email servers can be categorized into incoming and outgoing email servers.

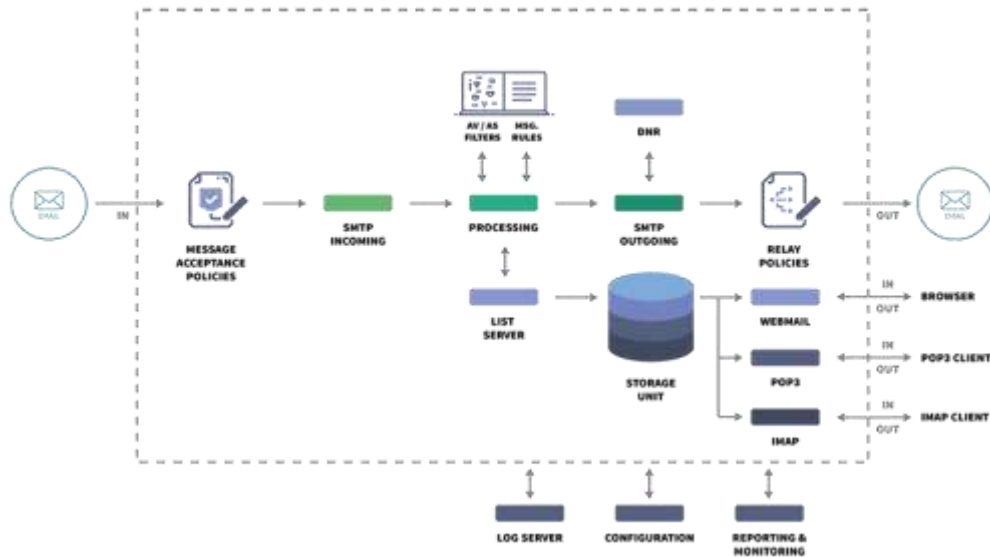


Figure 8:Email Server

For a computer system to function as a mail server, it must have **mail server software** installed. This software then allows the system administrator to manage and create email accounts for any of the domains hosted on the server.

Alongside that, you have **protocols**. Protocols are networking-software rules that allow computers to connect to networks everywhere so you may shop online, send emails, and browse the internet freely. These protocols are a vital component of your networking activity.

SMTP

The SMTP protocol handles any outgoing mail requests and sends emails. So SMTP is short for Simple Mail Transfer Protocol and is the **outgoing mail server**. We can think of SMTP as moving your email on and across networks. Without it, your emails wouldn't go anywhere.

POP / IMAP

For incoming mail servers, there are two main varieties — POP3 and IMAP. POP3 servers, short for Post Office Protocol version 3, are best known for fetching the content of the Inbox on your computer's hard drive. IMAP servers, short for Internet Message Access Protocol, are used for o way synchronization of the entire mailbox.

IMAP can also function as a mail server's alternate protocol retrieval program. And although there are newer POP versions, which offer more features, the preferred protocol remains POP3 because it's simple, has a high rate of success, and gets the

job done with a minimum of errors. It even allows you to download your emails and read them while offline.

Without one of these protocols working correctly, your email would not reach your computer.[12]

Results and Discussions :

This section presents the step-by-step approach and corresponding results for implementing the project requirements. Each configuration task was carried out systematically in Cisco Packet Tracer, and relevant screenshots are included to demonstrate the successful completion of the tasks. The IP address used in the project is obtained by the ID number of Aya Hamayl : 1221469 where $x=69 \rightarrow$ (Ex: 8.8.69.21)

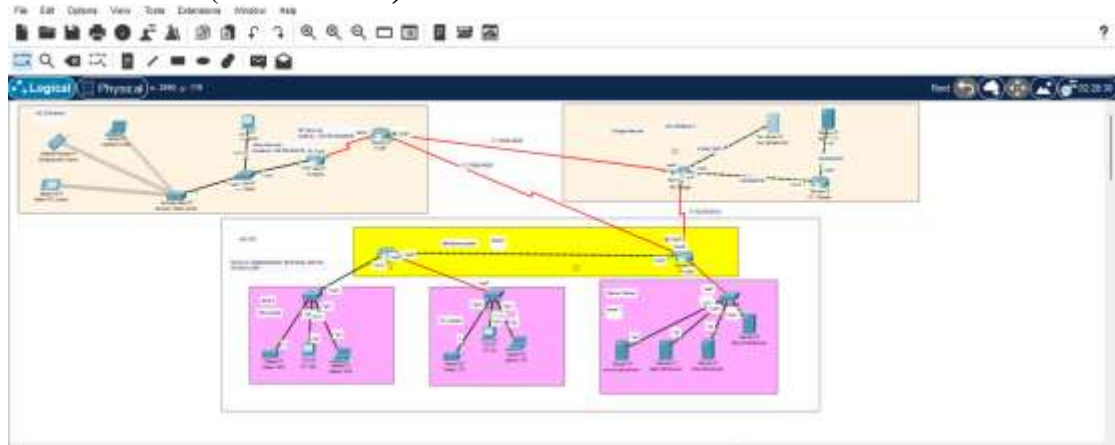


Figure 9 :main Topology

1. Google Network :

In These ASs we established a fully operational email server that supports sending and receiving emails within the Google Network. The static IP and domain name configuration ensure reliability, while user accounts provide secure access. Testing confirmed the system's functionality, verifying that it meets the project requirements.

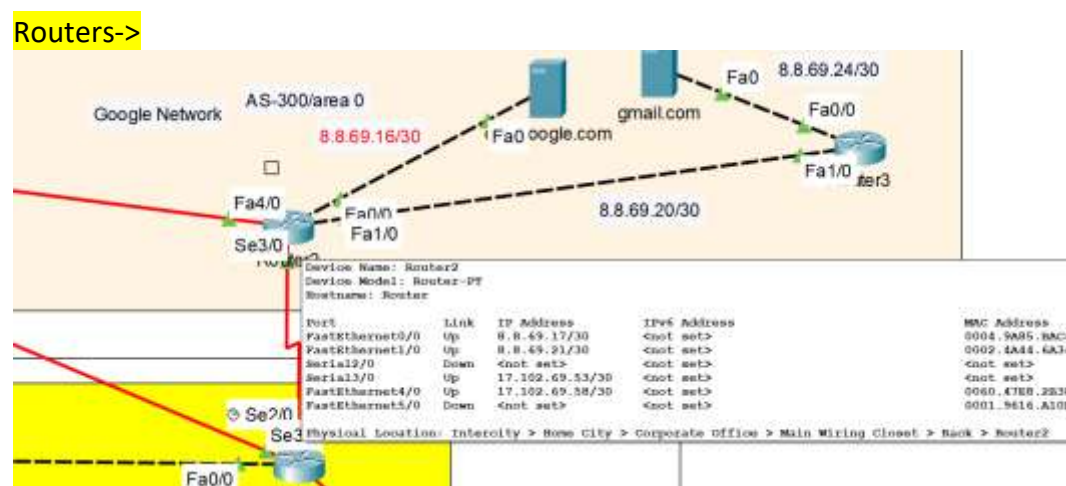


Figure 10:IP of R_google

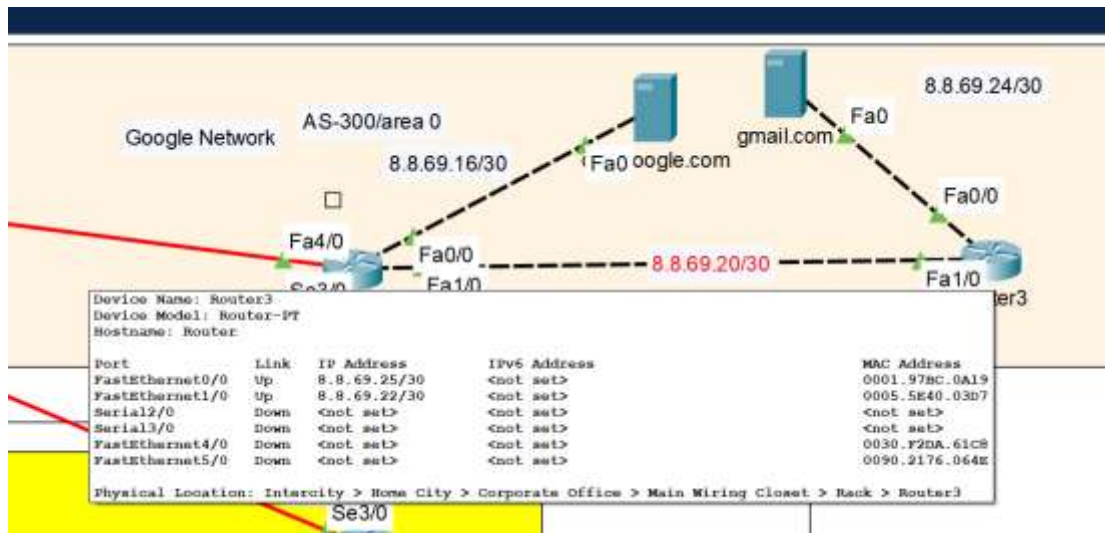


Figure 11: R3_google

dns.google.com ip →



Figure 12 :ip of dns

gmail.com ip→

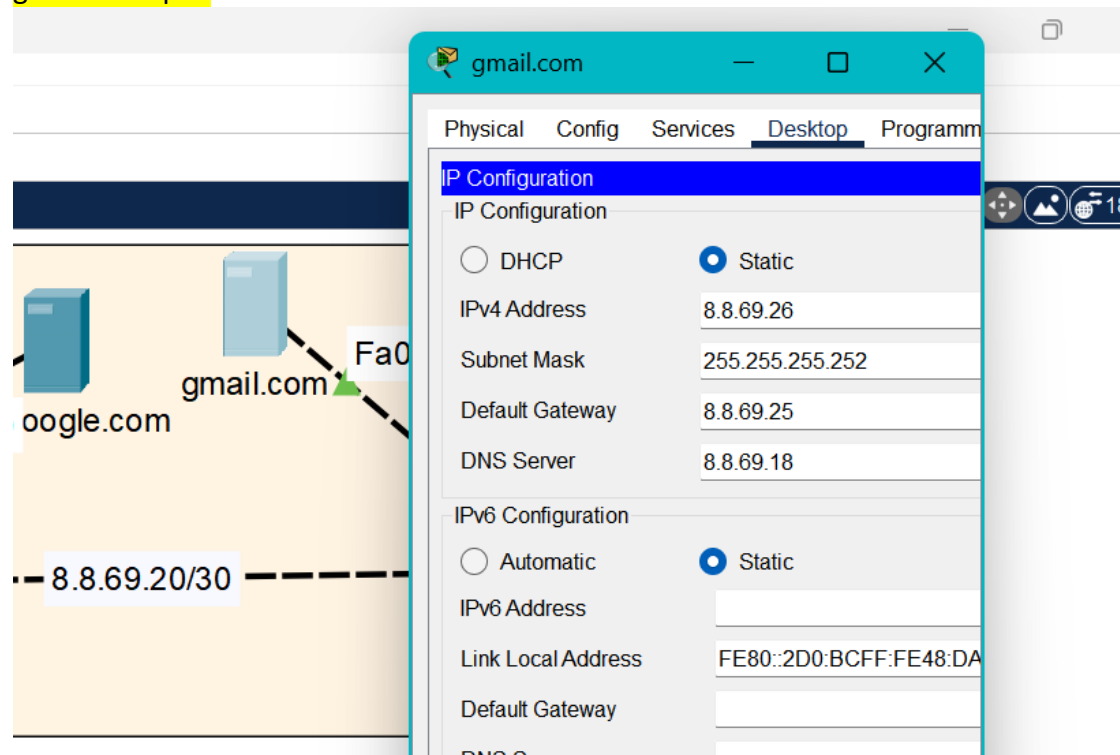


Figure 13:ip of gmail.com

2. Faculty of Engineering and Technology Network :

2.1. IP Subnetting :

we have a network consisting of 4 subnets. To determine the subnets, Since the IP is 180.69.72.0/24.

4 Splits with different sizes are Required :

- Servers Subnet: Requires 28 IP addresses.
- Electrical and Computer Engineering (ECE) Subnet: Requires 120 IP addresses.
- Computer Science (CS) Subnet: Requires 55 IP addresses.
- Backbone Subnet: Requires 4 IP addresses.

Main IP: 180.69.72.00000000/24

Take the first Client Bit from the left and add it to the Network Bits:

180.27.72.00000000

Two possibilities 0 and 1:

180.27.70.00000000 -> 180.69.72.0/25 → ECE Subnet

180.27.70.10000000 -> 180.69.72.128/25

(128 Available IP with network and Broadcast IP)

Next IP Used: 180.27.70.10000000

180.27.70.10000000 -> 180.69.72.128/26 → CS Subnet

180.27.70.11000000 -> 180.69.72.192/26

(64 Available IP with network and Broadcast IP)

Next IP Used: 180.27.70.11000000

180.27.70.11000000 -> 180.69.72.192/27 → Servers Subnet

180.27.70.11100000 -> 180.69.72.224/27

(32 Available IP with network and Broadcast IP)

Next IP Used : 180.27.70.11100000

180.27.70.11100000 -> 180.69.72.224/28 → Backbone Subnet

(16 Available IP with network and Broadcast IP)

The Table below shows the IP Addresses Details :

Table1 : IP addresses

Subnet	IP Range	CIDR	Subnet Mask	Broadcast IP	First Usable Host IP	Last Usable Host IP	Required IP Addresses
Electrical and Computer Engineering (ECE)	180.69.72.0 - 180.69.72.127	180.69.72.0/25	255.255.255.128	180.69.72.127	180.69.72.1	180.69.72.126	120
Computer Science (CS)	180.69.72.128 - 180.69.72.191	180.69.72.128/26	255.255.255.192	180.69.72.191	180.69.72.129	180.69.72.190	55
Servers	180.69.72.192 - 180.69.72.223	180.69.72.192/27	255.255.255.224	180.69.72.223	180.69.72.193	180.69.72.222	28
Backbone	180.69.72.224 - 180.69.72.227	180.69.72.224/30	255.255.255.252	180.69.72.227	180.69.72.225	180.69.72.226	4

2.2 Static IP configuration for Printer_ECE and Printer_CS. :

Printer_ECE with static IP → 180.69.72.1

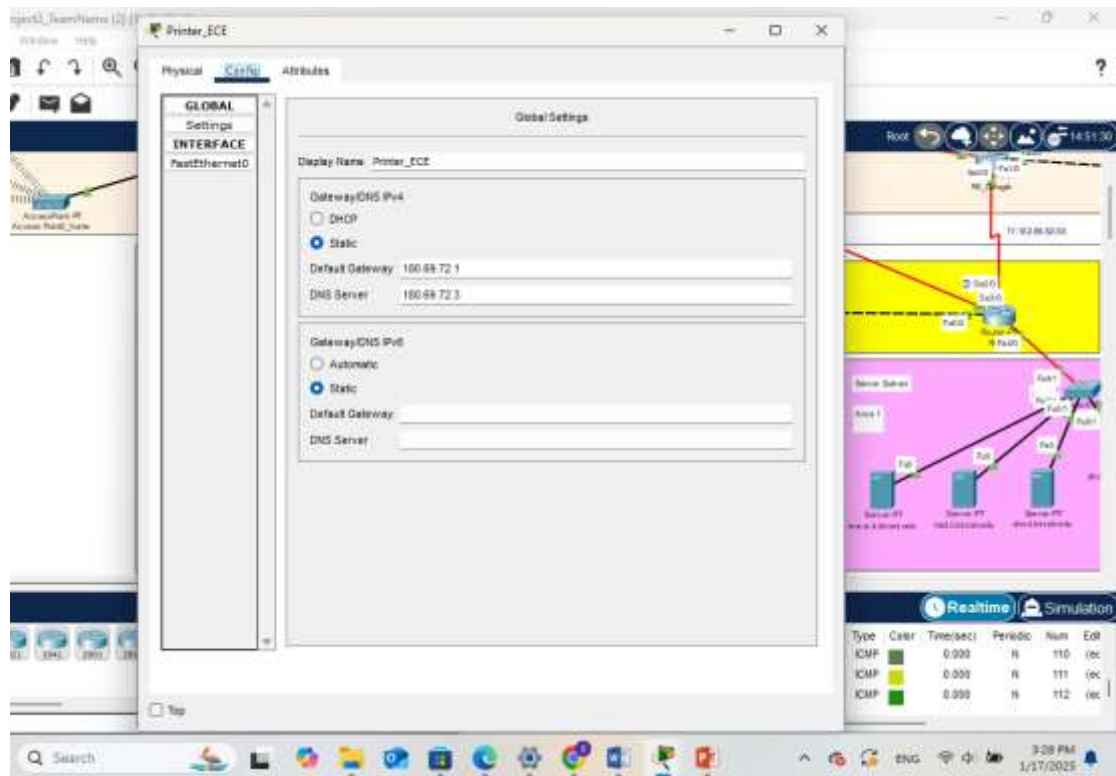


Figure 14 :printer_ECE config

Printer_CS with static ip → 180.69.72.129

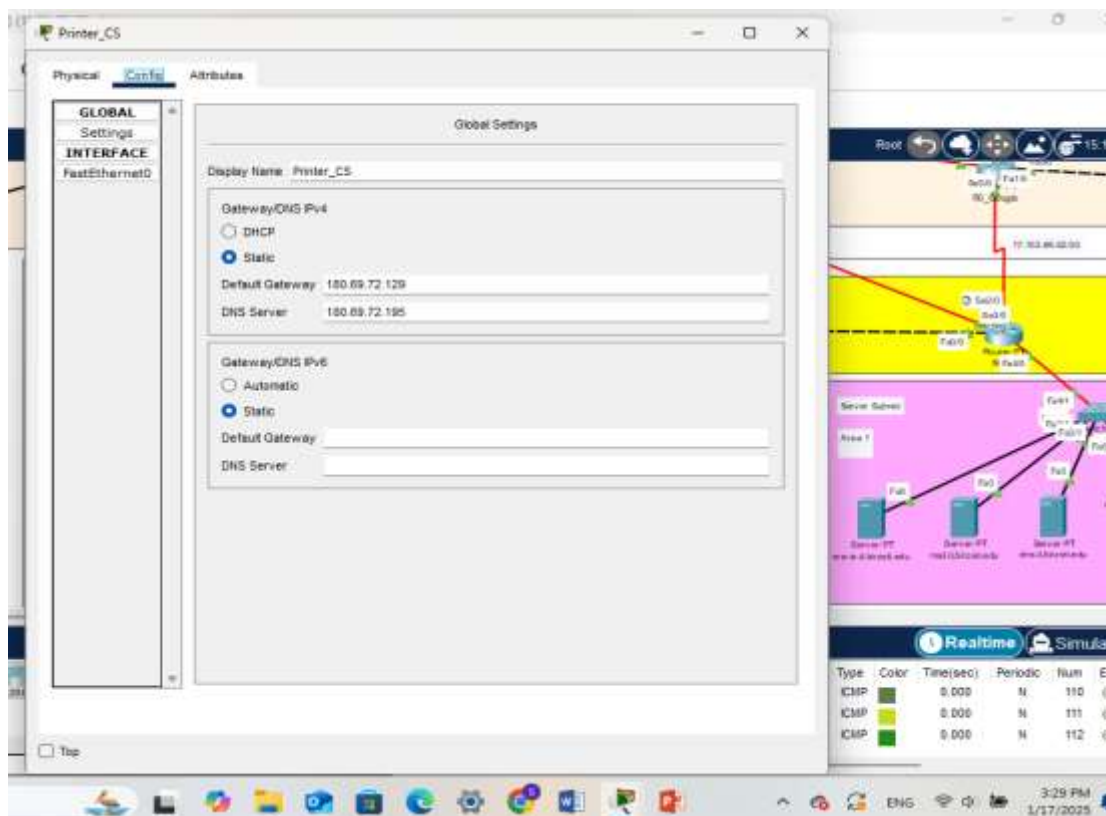


Figure 15 :printer_CS

2.3 Dynamic IP configuration for an ECE network end device and a CS network end device.

ECE :

PC_ECE with dynamic ip → 180.69.72.11 :

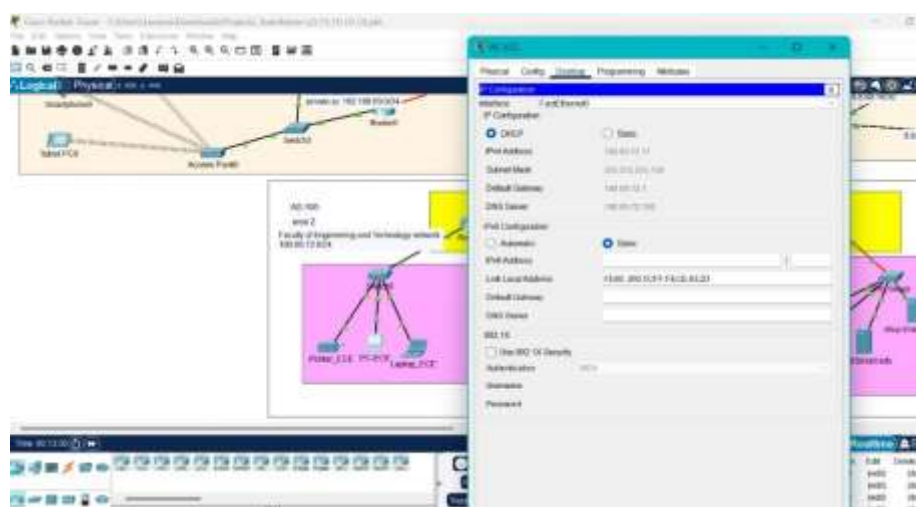


Figure 16: PC_ECE IP

Laptop_CS with dynamic ip → 180.69.72.12 :



Figure 17 :Laptop_CS IP

In CS Network :
PC_CS with dynamic ip → 180.69.72.139 :

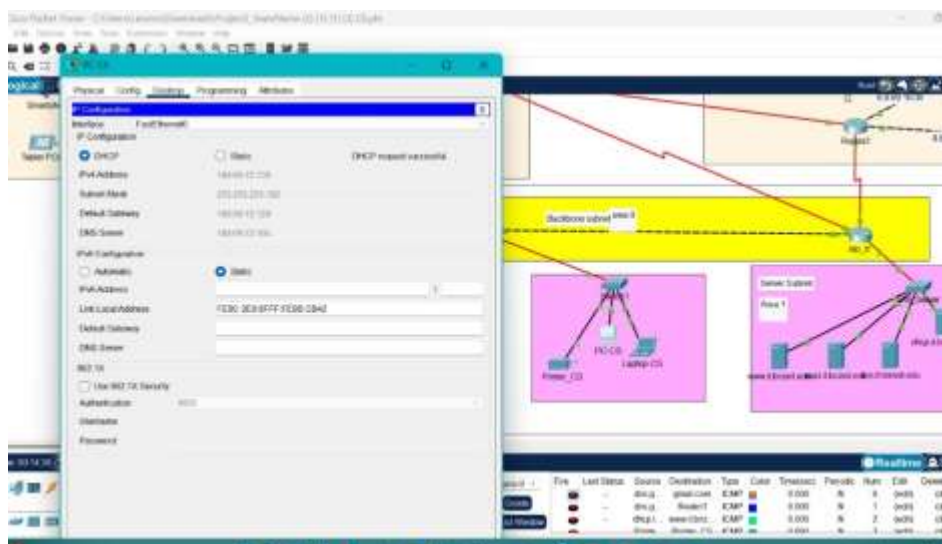


Figure 18 :PC_ECE IP

Laptop_ECE with dynamic ip → 180.69.72.140 :

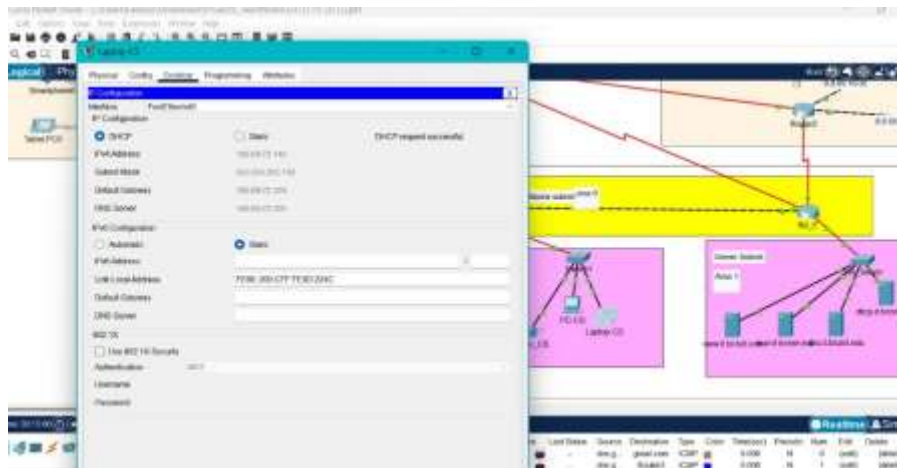


Figure 19 : Laptop_ECE IP

2.4 Static IP configuration for an R0_IT router port and an R1_IT router port.

R1_IT :

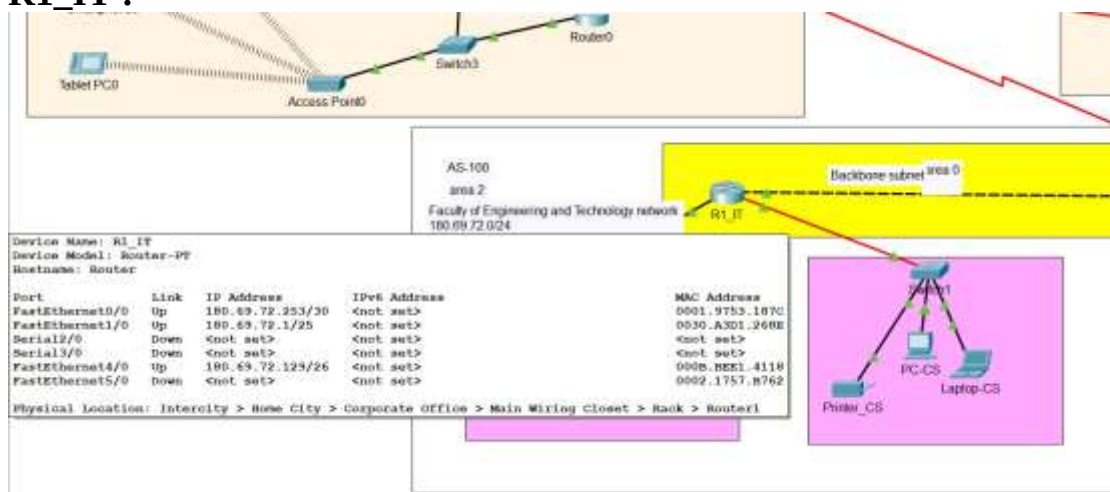


Figure 20 :R1_IT IP

R0_IT :

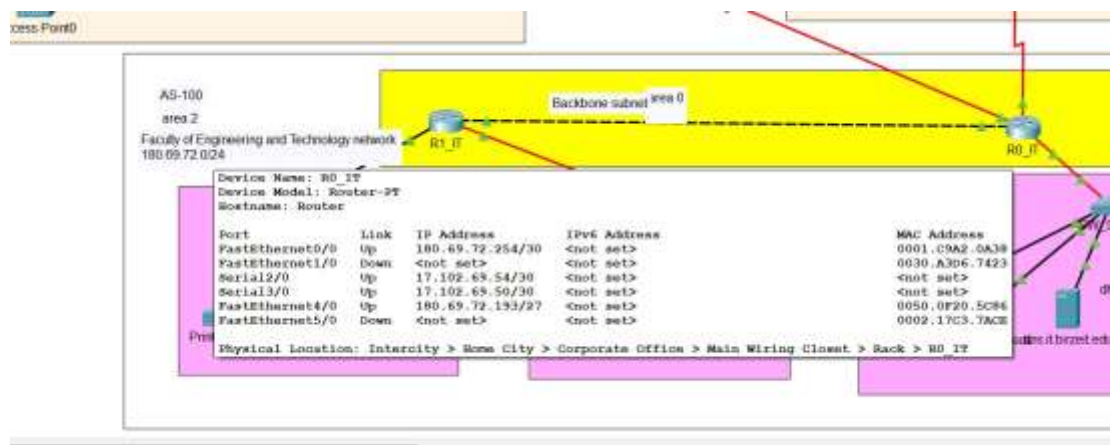
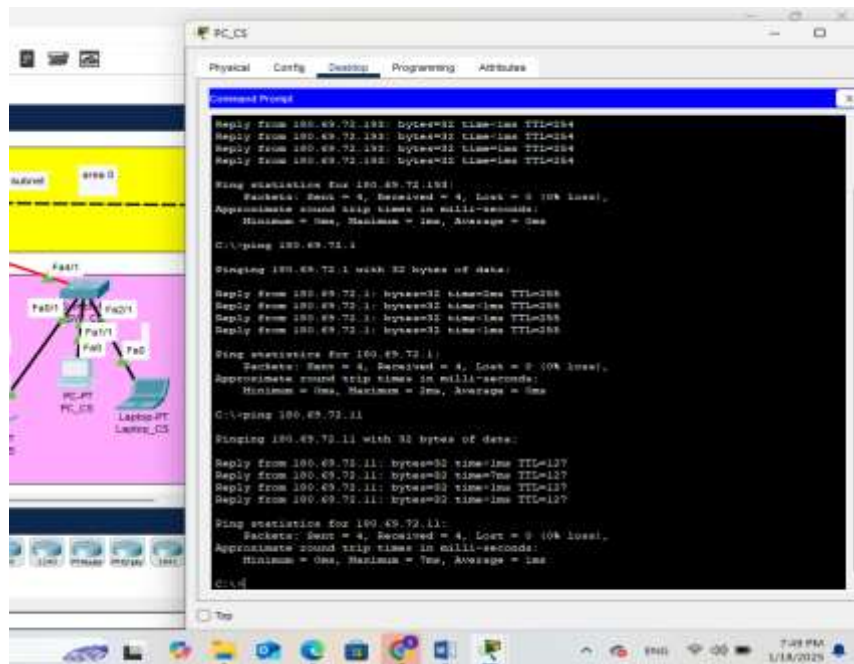
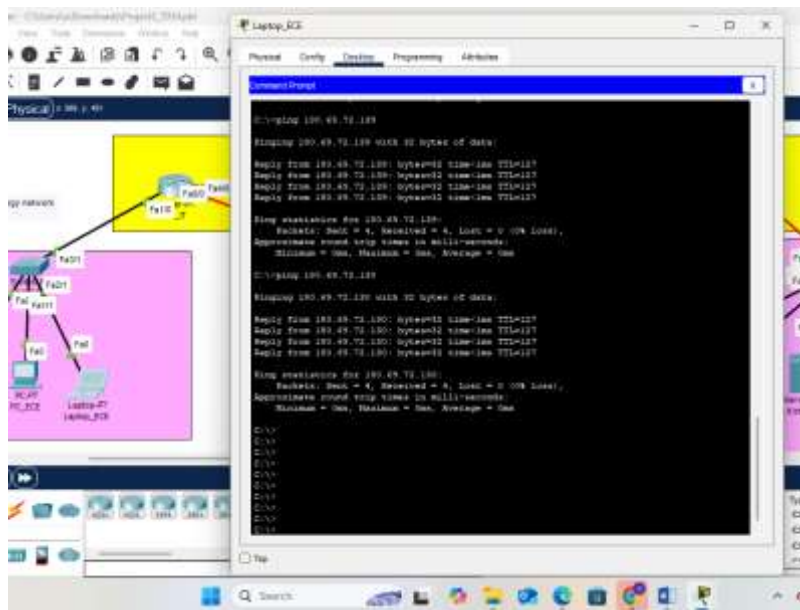


Figure 21 : R0_IT IP

2.5 : Successful Ping results within the Faculty of Engineering and Technology network. :

As shown in the picture we successfully make ping from any device in Faculty of Engineering and Technology network. :

From the below pictures, we try to ping to from 2 different devices and sending 5 packets and we can see that all packets arrive without and lost packet or requested out.



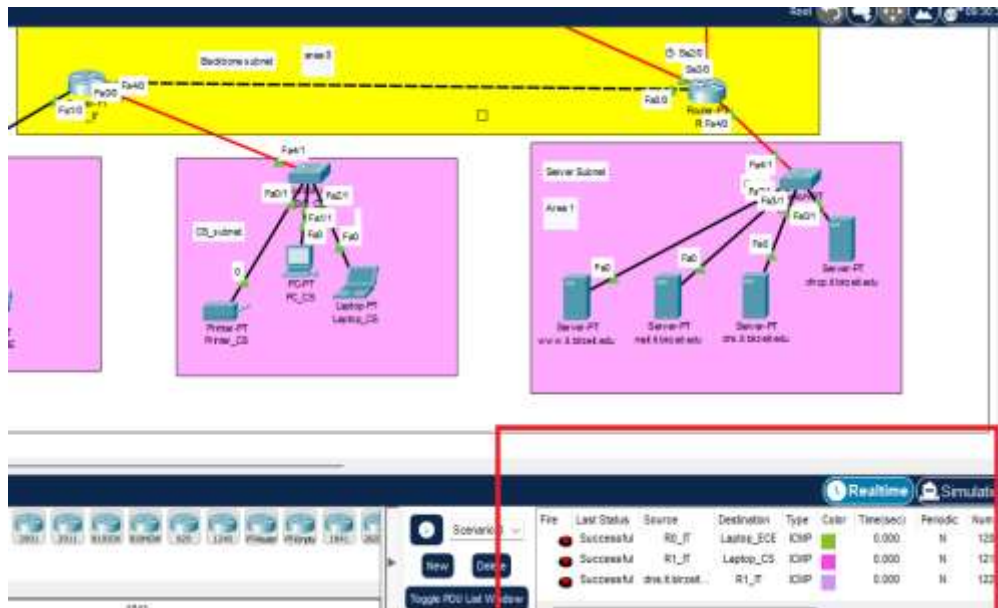


Figure 24 : PING 3

2.5 :IP configuration of Web, Email, DNS, and DHCP servers :

According to tabel below the server subnet have the range ip from 180.69.72.192→ 180.69.72.223 so from range of these IP addresses we distributed to this serer :

2.5.1 : IP configuration of Web :

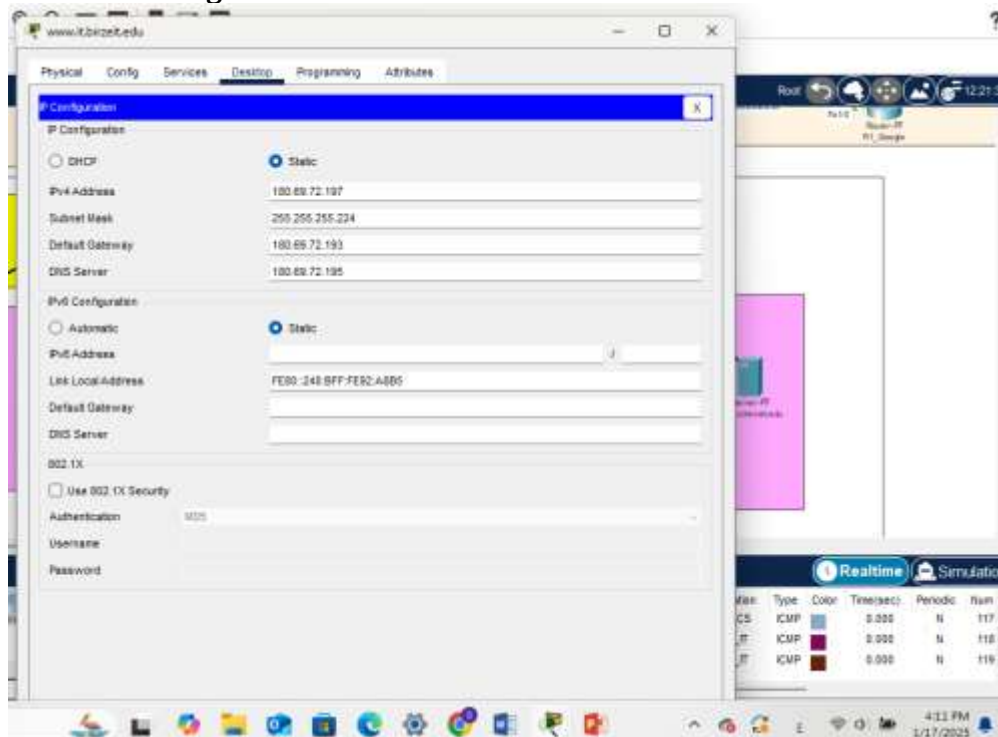


Figure 25 : IP of web

2.5.2 : IP configuration of Email :

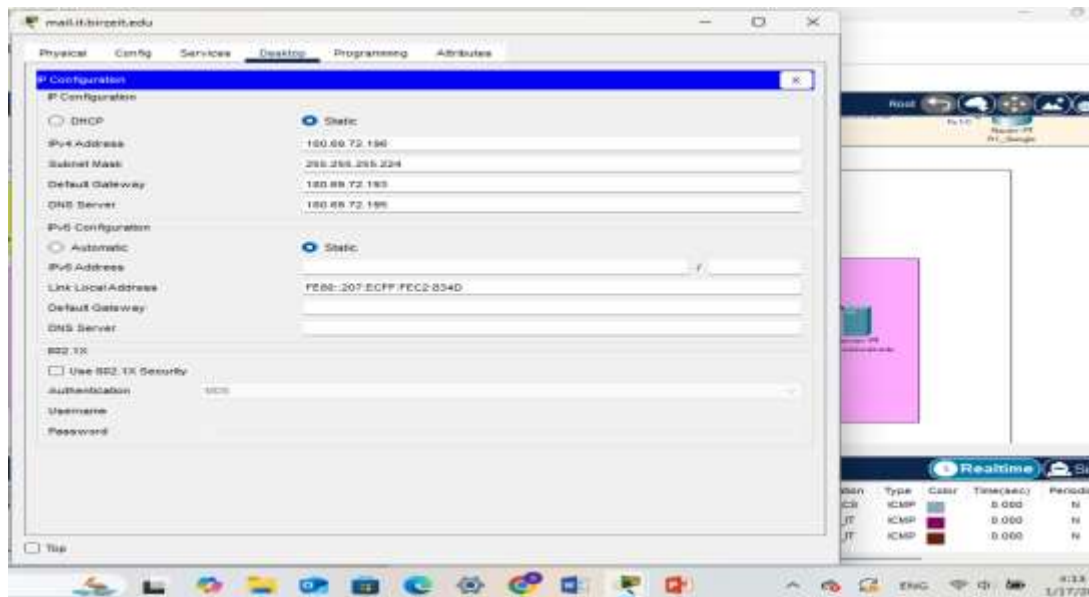


Figure 26 : IP of email

2.5.3 : IP configuration of DNS :

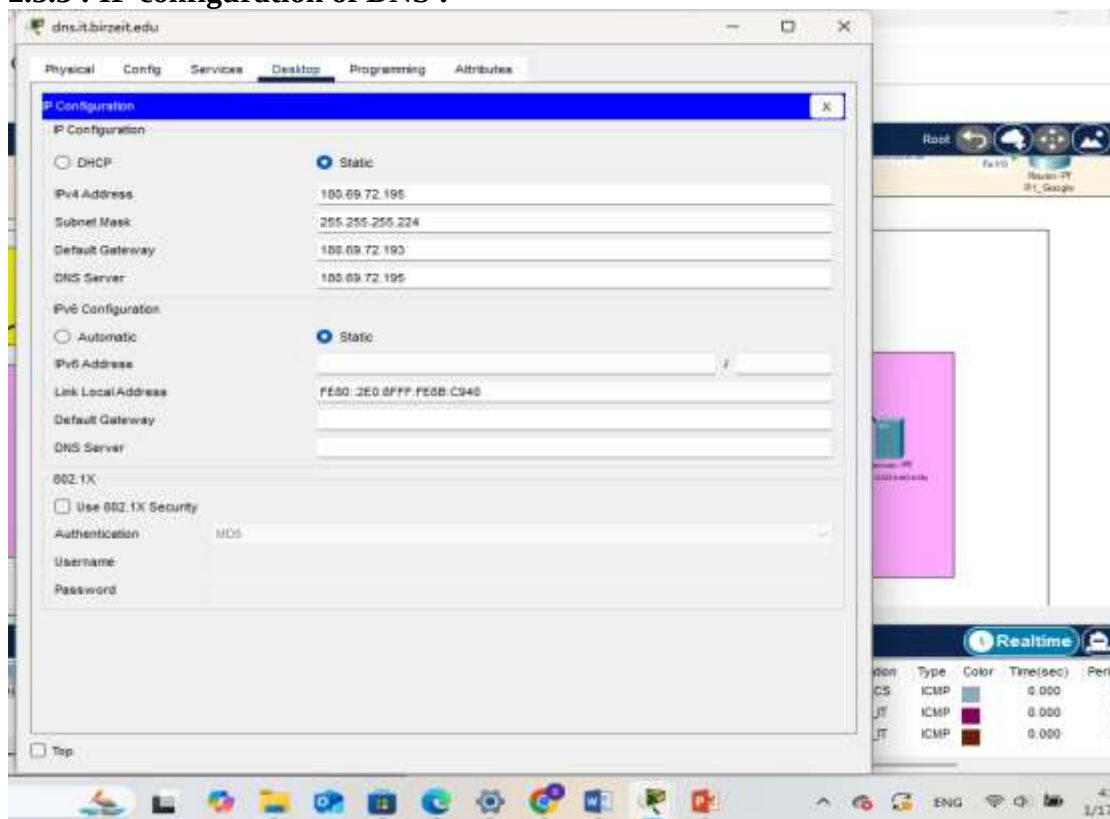


Figure 27 : IP of DNS

2.5.3 : IP configuration of DHCP :

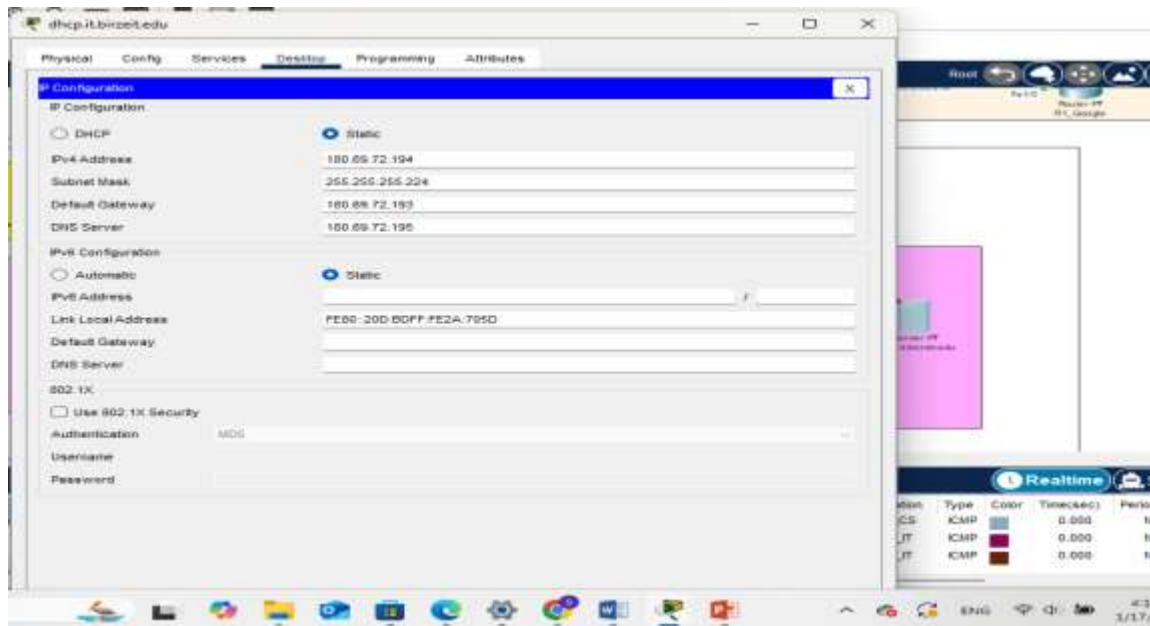


Figure 28 : IP of DHCP

2.6 : Email service with the user setup on the mail.it.birzeit.edu :

In this mail server we turn on the SMTP and POP3 server to send and receive the email we add two email address will follow this format : SID@it.birzeit.edu 1220481@it.birzeit.edu & 1221469@it.birzeit.edu

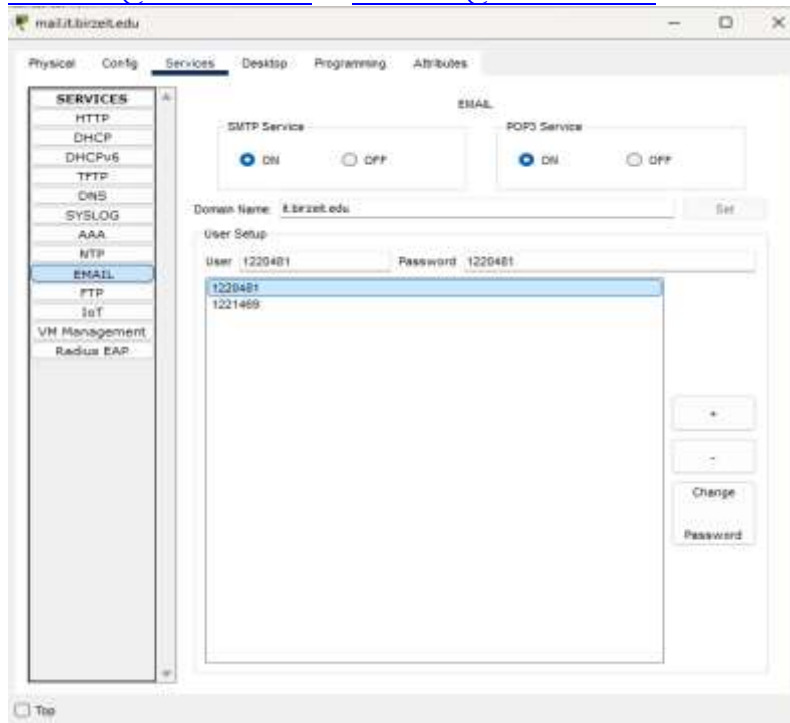


Figure 29 : first email address

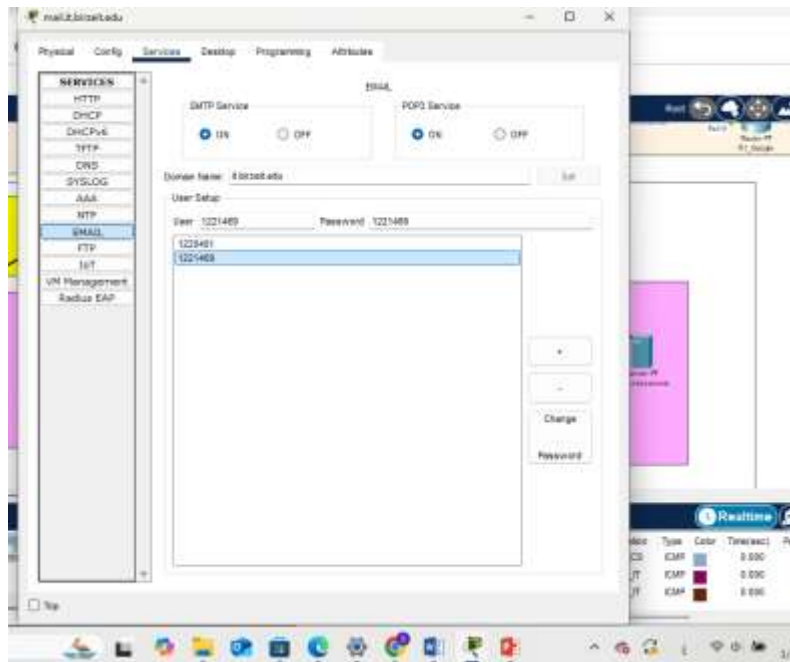


Figure 30 : second email address

2.7 DNS service with the RRs on the dns.it.birzeit.edu :

In this step we turn on the dns server and add requerd recourse records

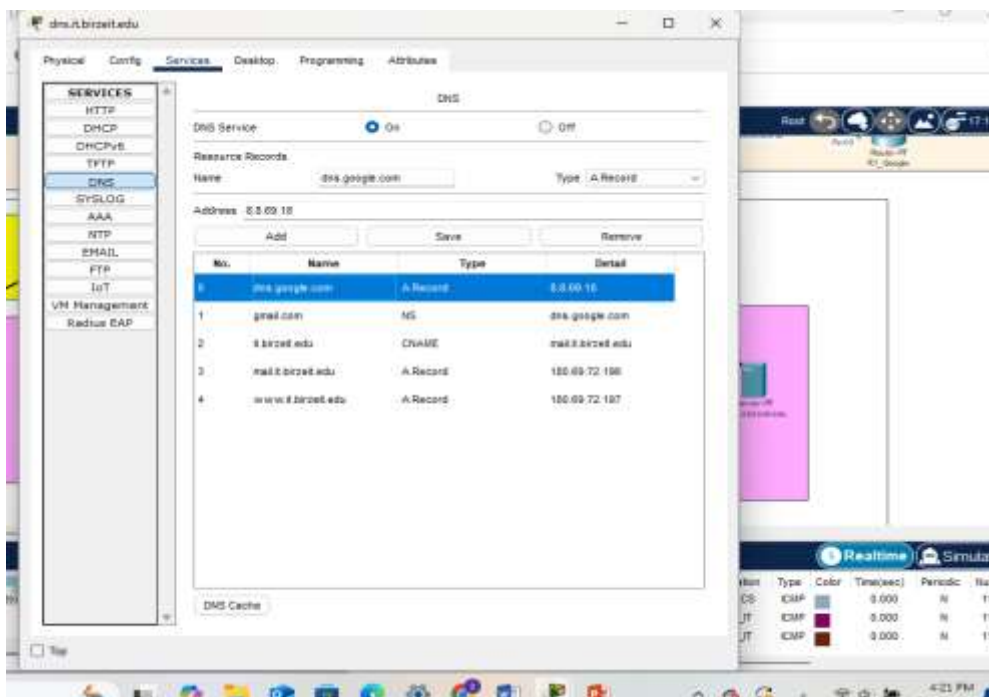


Figure 31 : DNS With RRS 1

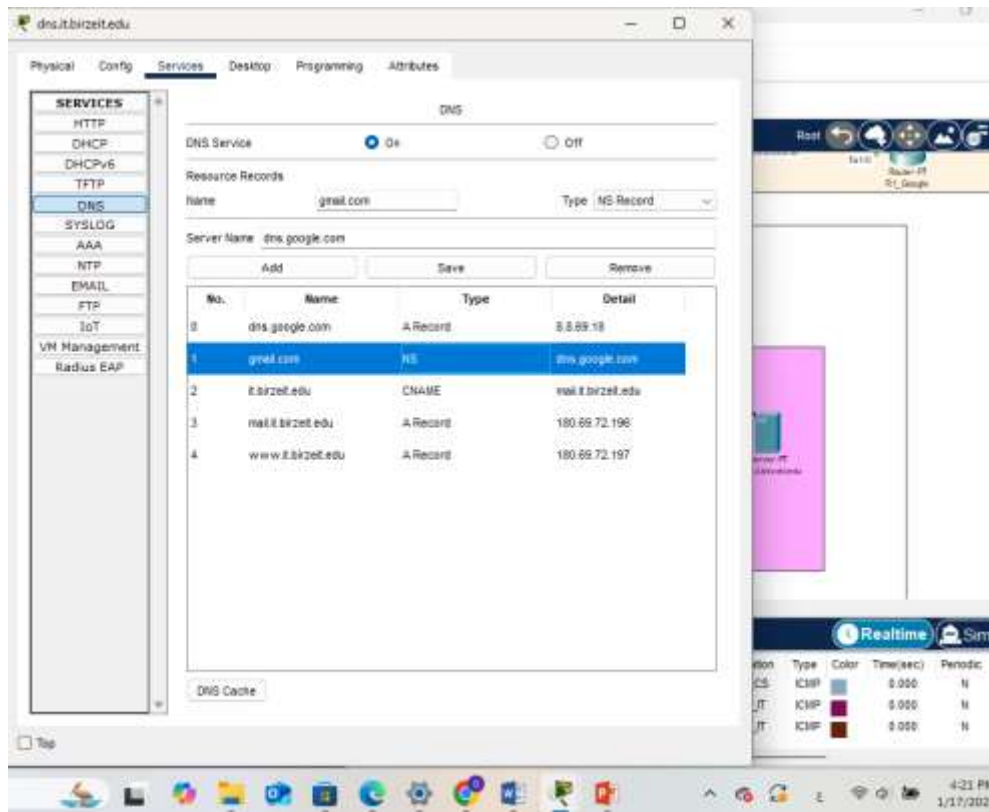


Figure 32: DNS With RRS 2

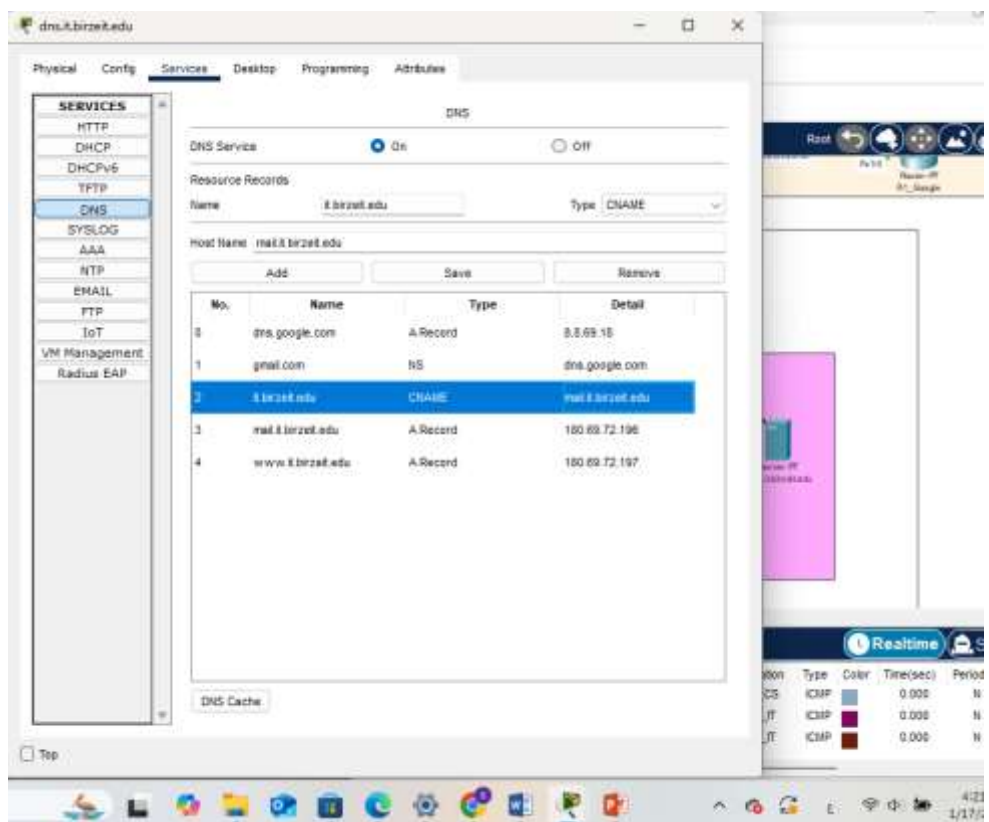


Figure 33 : DNS with RRs 3

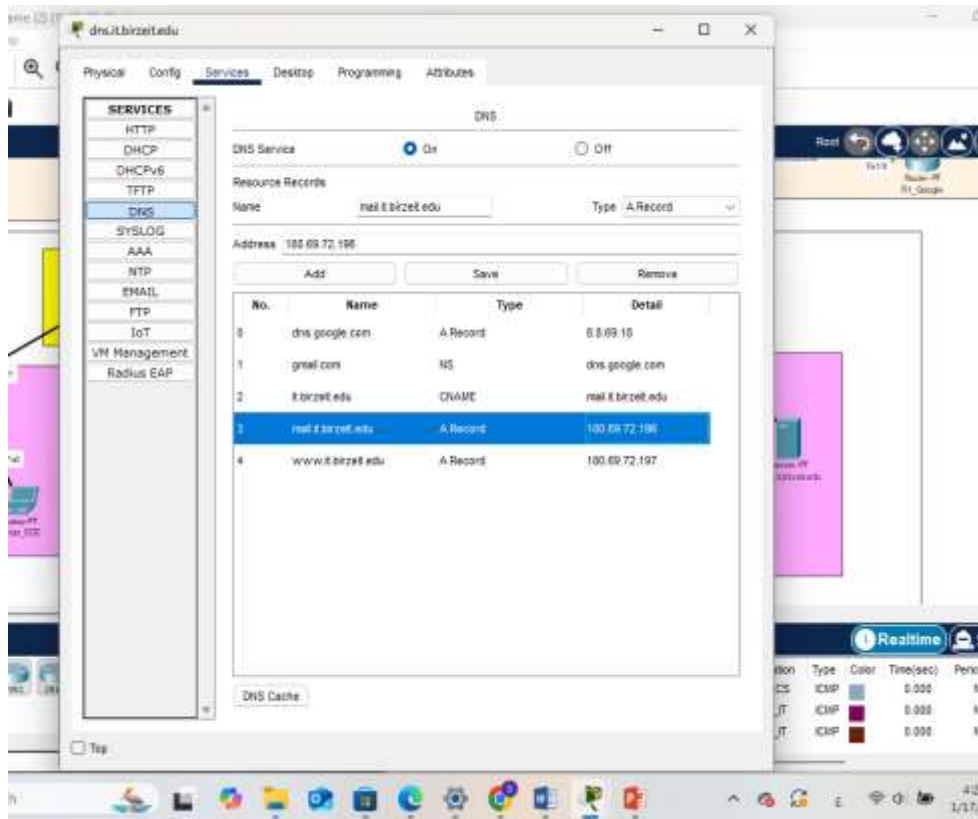


Figure 34 :DNS with RR 4

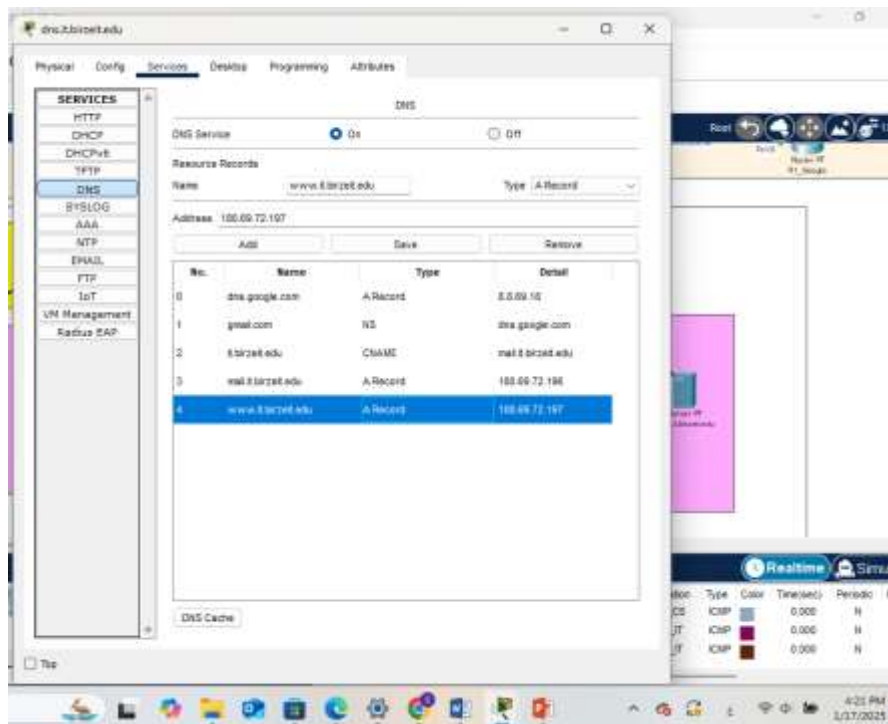


Figure 35 :DNS with RR 5

2.8 DHCP service with the pools on the dhcp.it.birzeit.edu. edu :

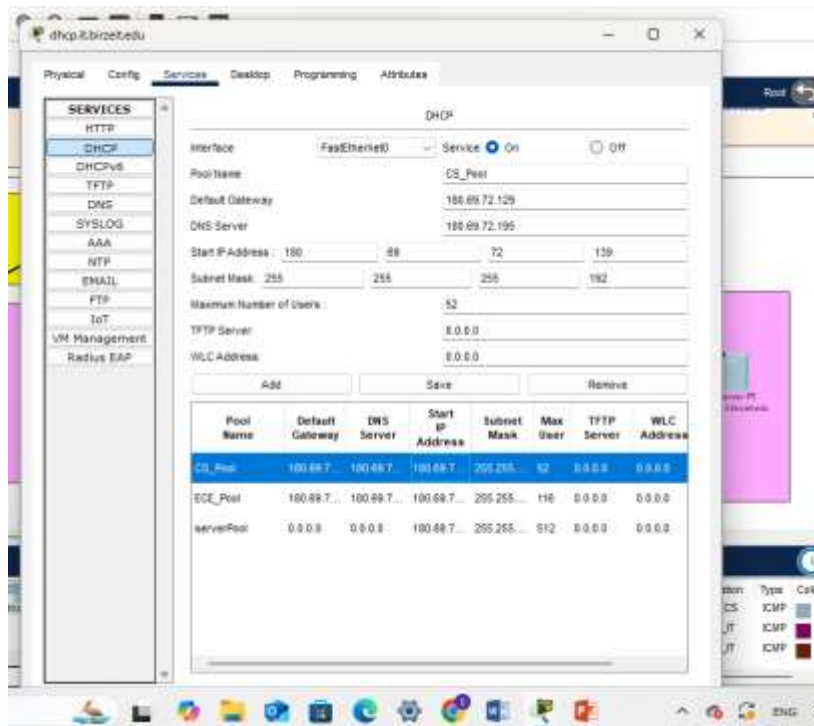


Figure 36 :DHCP pool 1

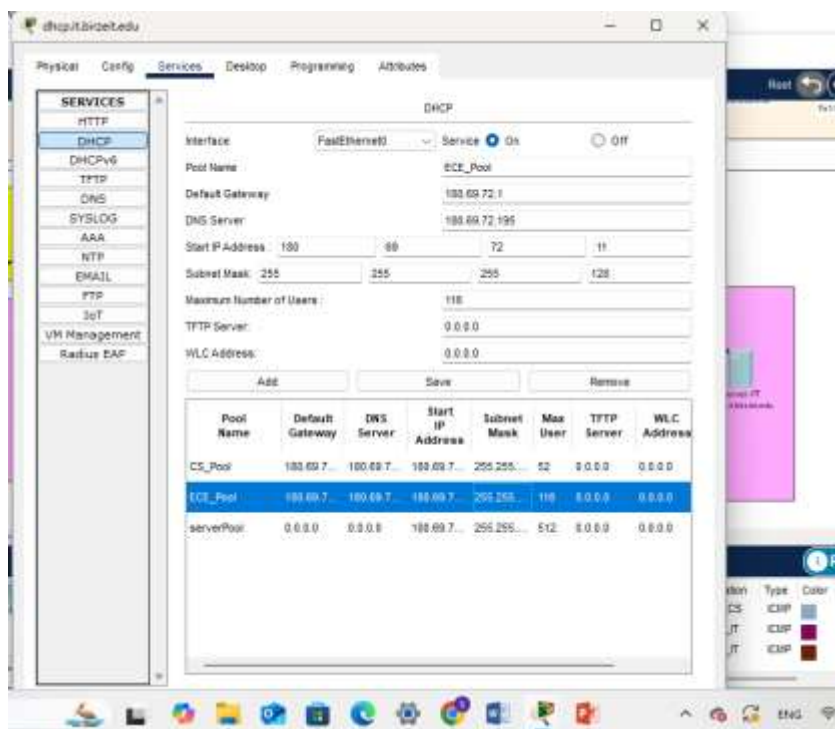


Figure 37 : DHCP pool 2

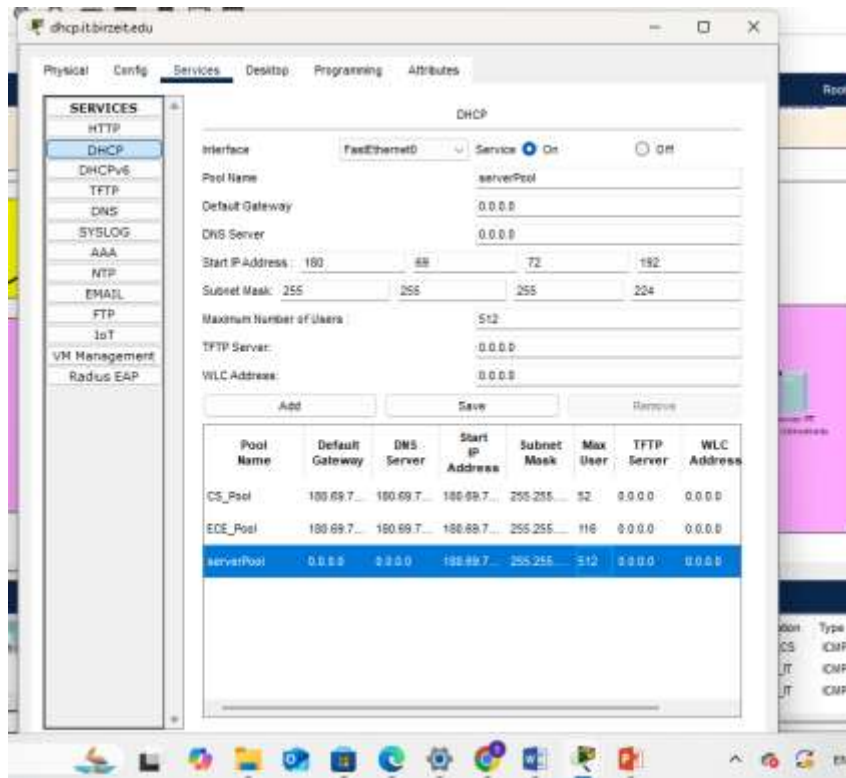


Figure 38 : DHCP pool 3

2.9 Successful access to www.it.birzeit.edu from a Faculty of Engineering and Technology network end device. :

From Laptop_CS:

We access a domain from a device, open the device and select “Desktop” then Web browser and enter the domain www.it.birzeit.edu

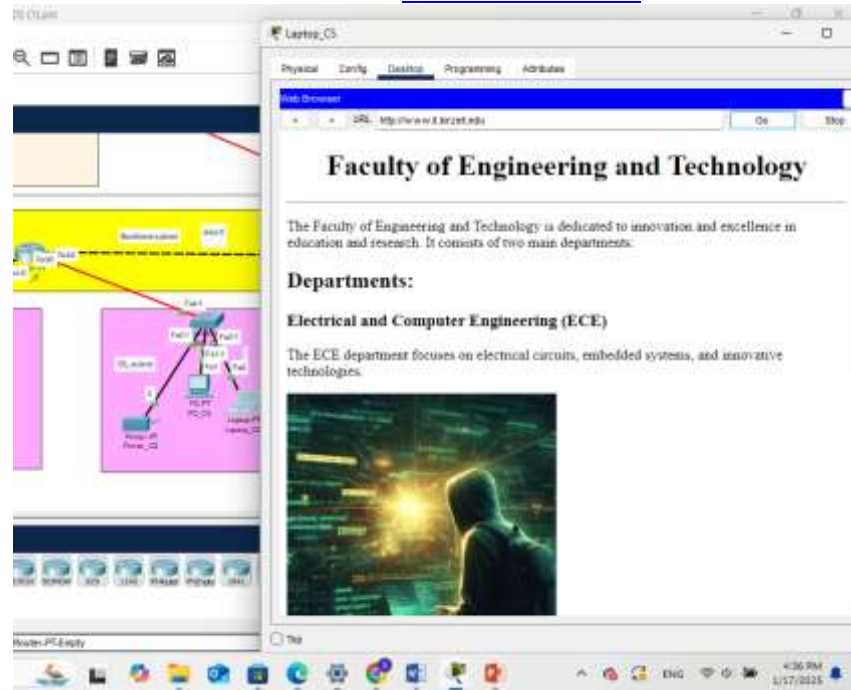


Figure 39 : Open www.it.birzeit.edu from laptop_cs

From PC_CS :



Figure 40 :Open www.it.birzeit.edu from PC_CS

From laptop_ECE :

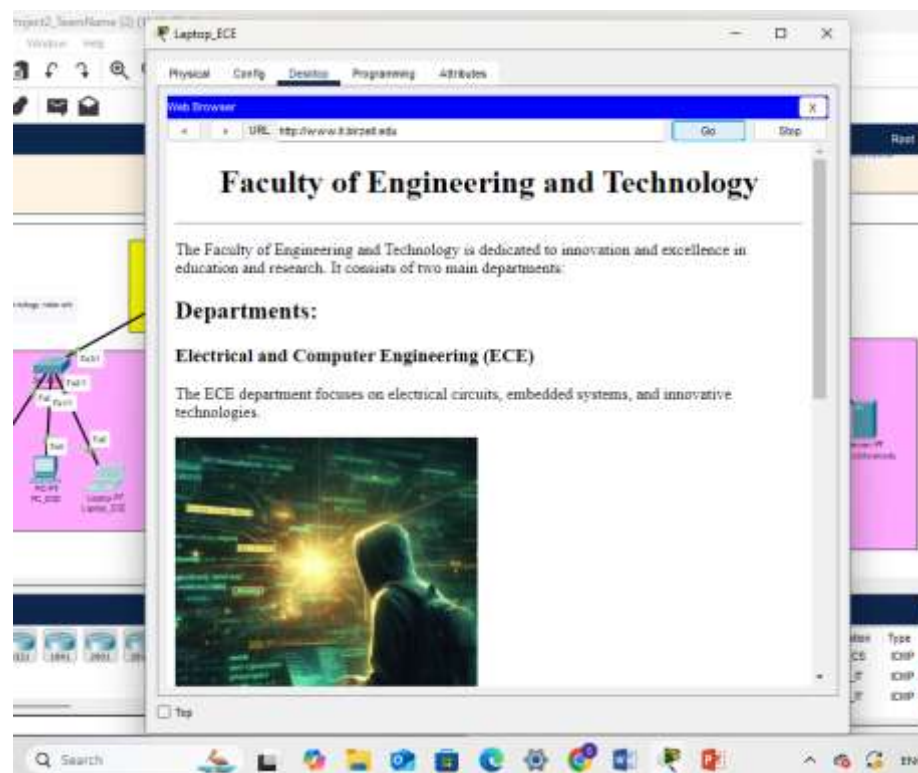


Figure 41 :Open www.it.birzeit.edu from laptop_ECE

From PC_ECE :

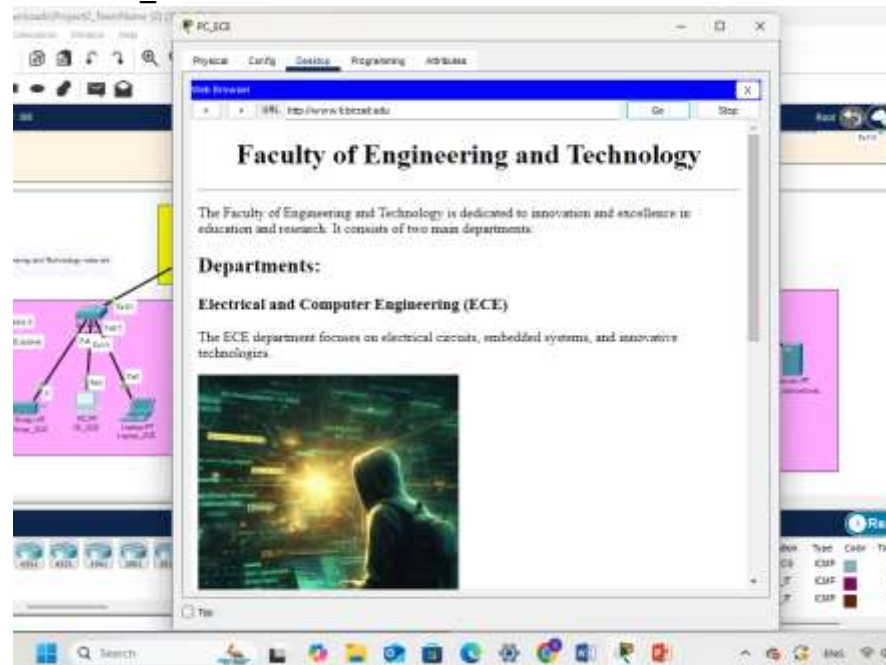


Figure 42 :Open www.it.birzeit.edu from PC_ECE

2.10 Email client configuration for gmail.com and it.birzeit.edu accounts

2.10.1 client for gmail.com in PC_CS & PC_ECE : in all PC device we Configure the email client

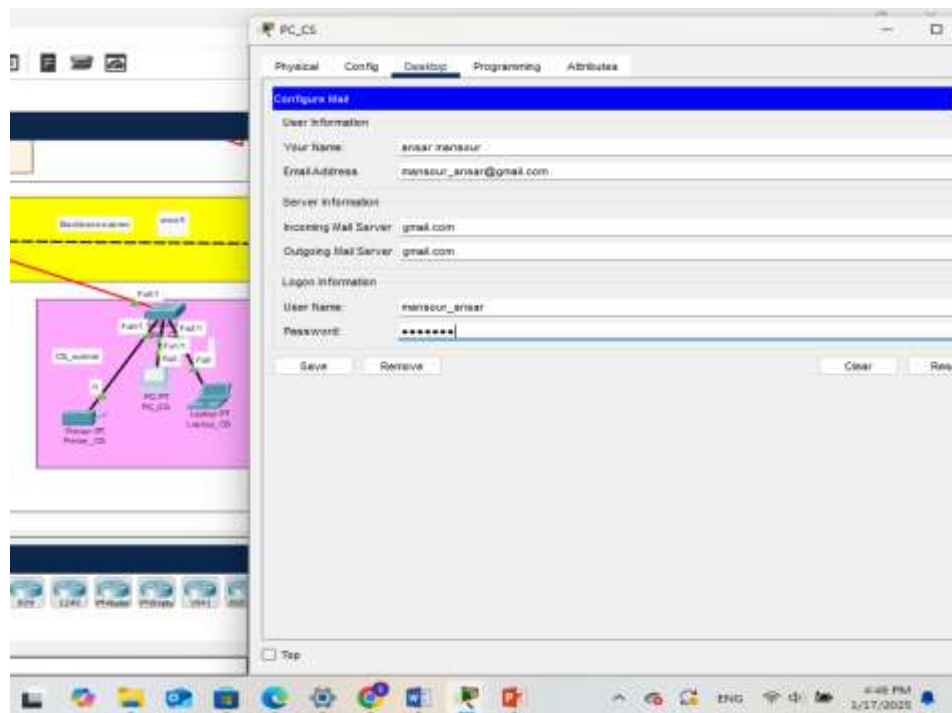


Figure 43 : Gmail client in PC_CS

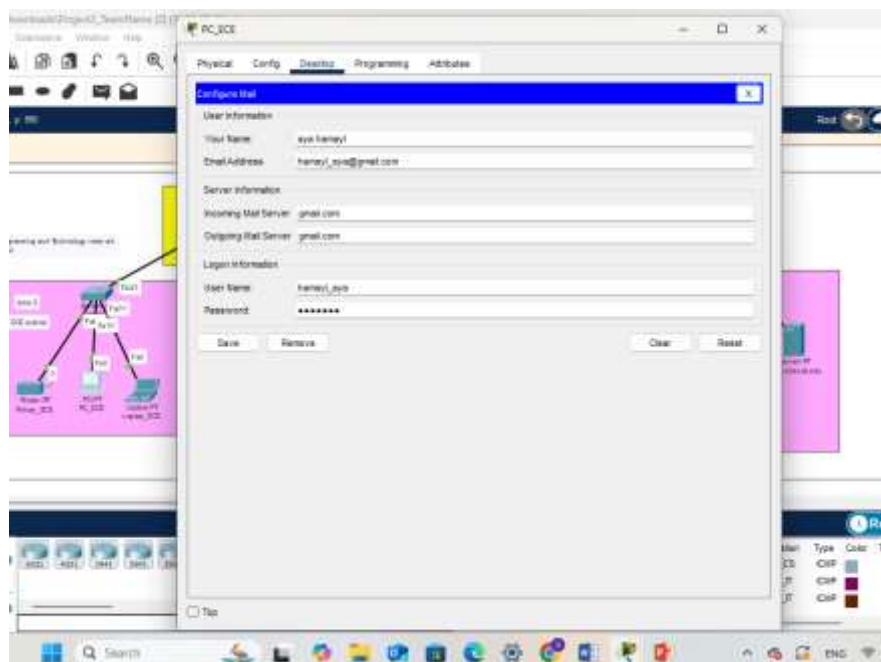


Figure 44 : Gmail client in PC_ECE

2.10.2 client for it.birzeit.edu in Laptop_CS & Laptop_ECE : in all PC device we Configure the it.birzeit.edu client

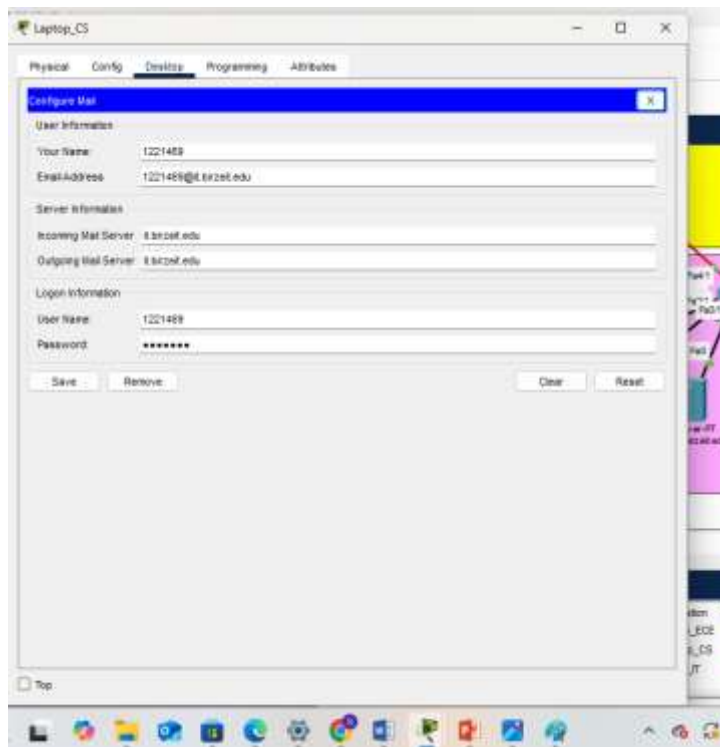


Figure 45 :it.birzeit.edu client in Laptop_CS

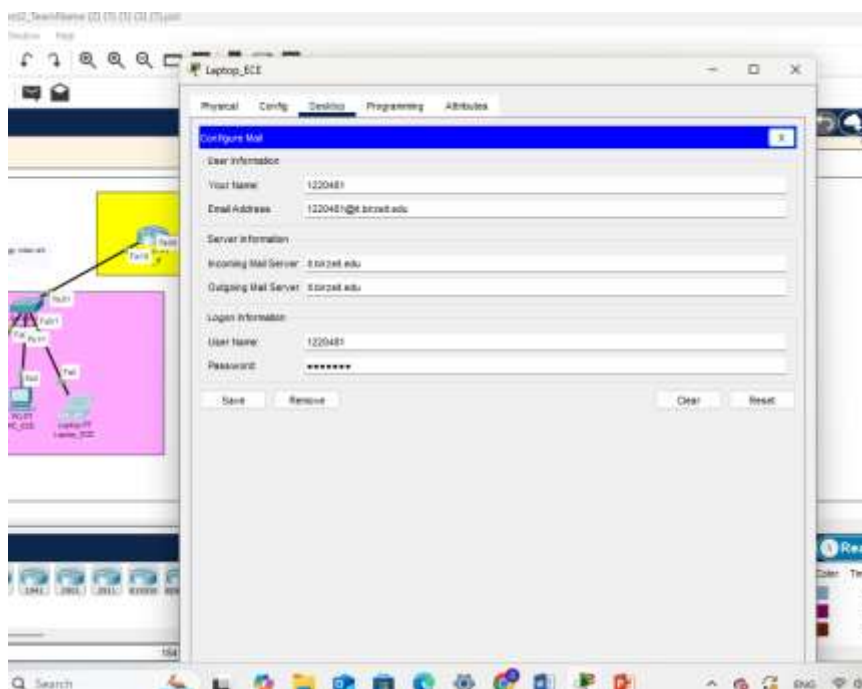


Figure 46 : it.birzeit.edu client in Laptop_ECE

2.11 : Successful sending and receiving of emails: Between Gmail accounts within the Faculty of Engineering and Technology network. & Between Birzeit accounts within the Faculty of Engineering and Technology network

2.11.1 : we send email from mansour_ansar@gmail.com in PC_CS to hamayl_aya@gmail.com in PC_ECE :

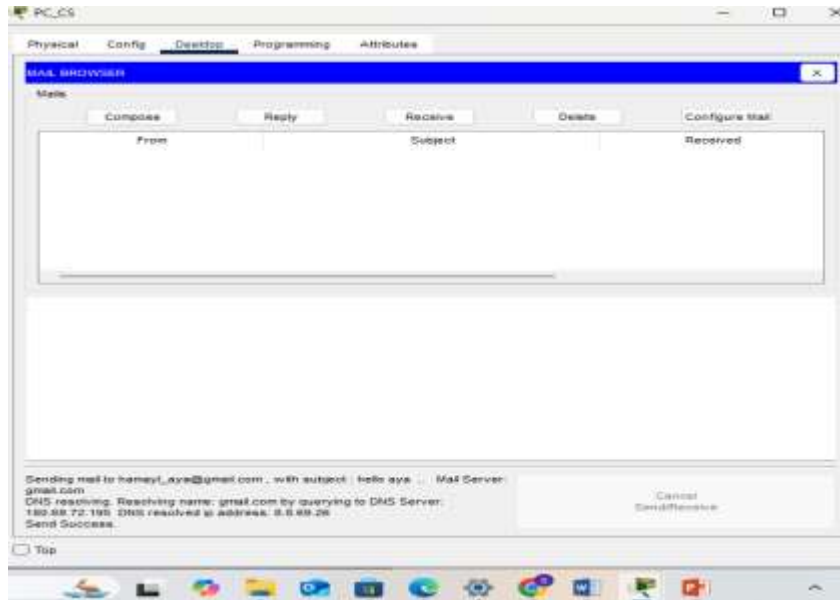


Figure 47 :send email to another gmail client

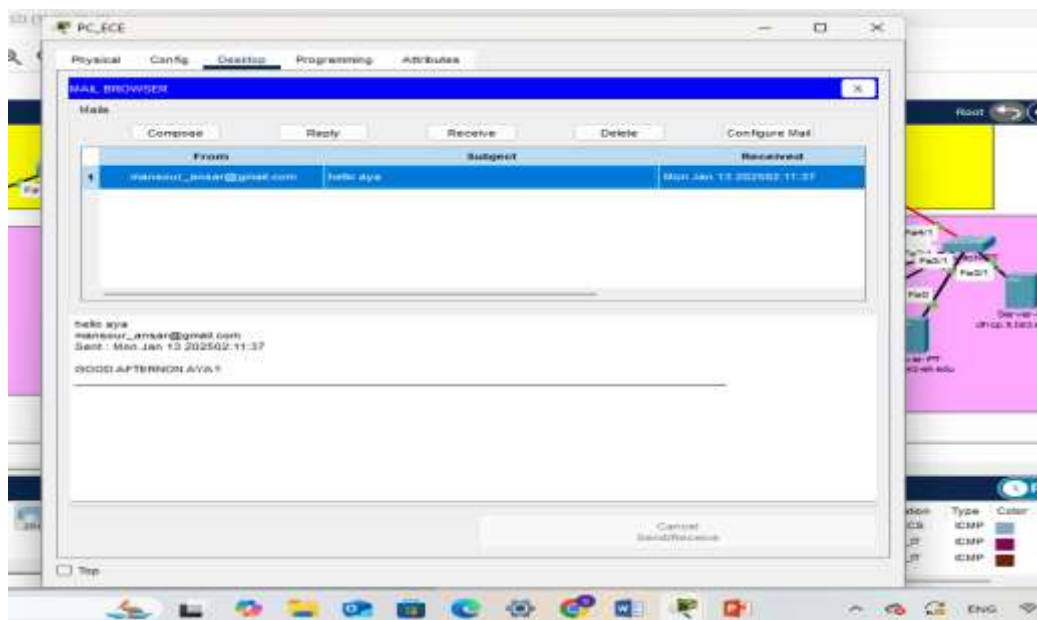


Figure 48 : resived email from another gmail client

2.11.2 : we send email from in PC_ECE to hamayl_aya@gmail.com mansour_ansar@gmail.com in PC_CS :

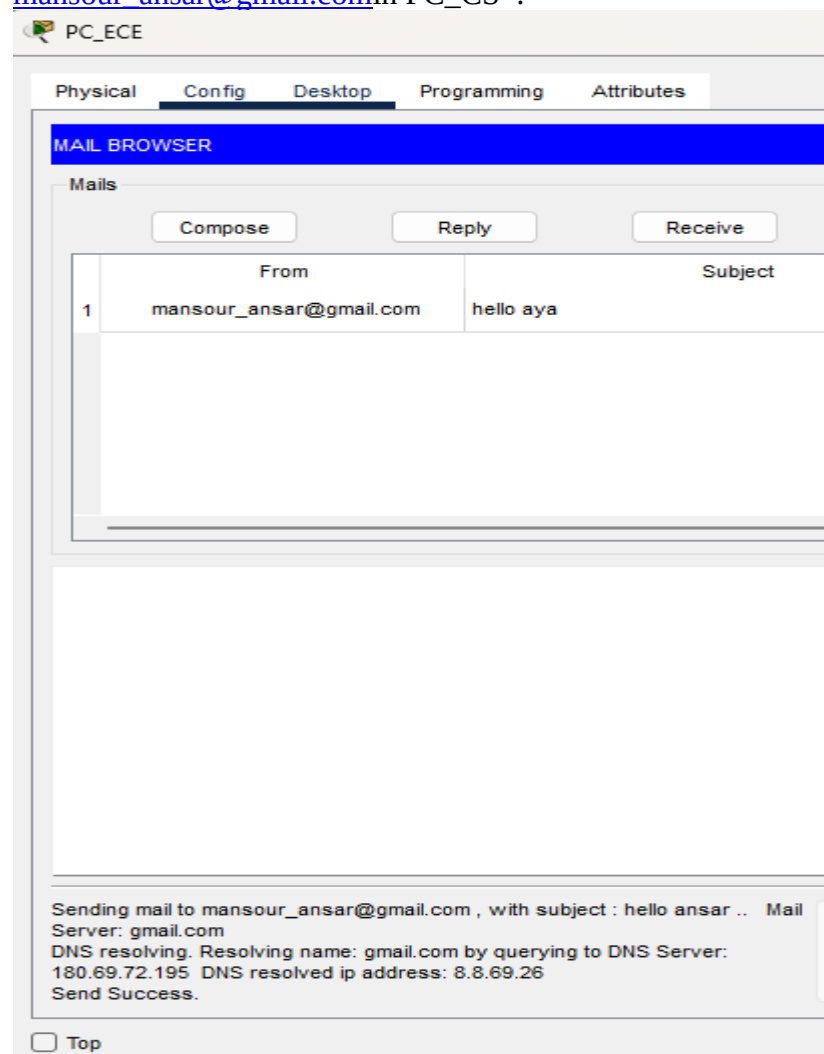


Figure 49 : send email to another gmail client

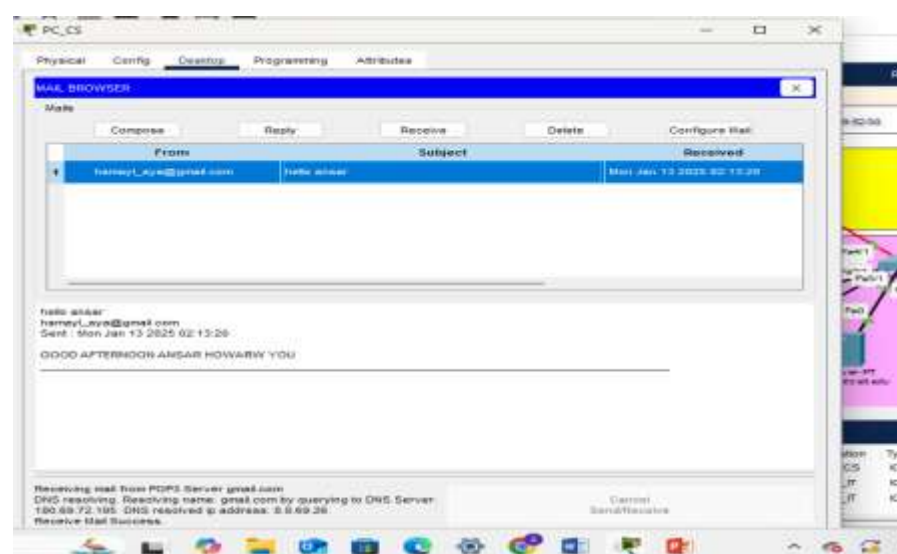


Figure 50 : resived email frome another gmail client

2.11.3: we send email from 1220481@it.birzeit.edu in laptop_ECE to 1221469@it.birzeit.edu in Laptop_CS :

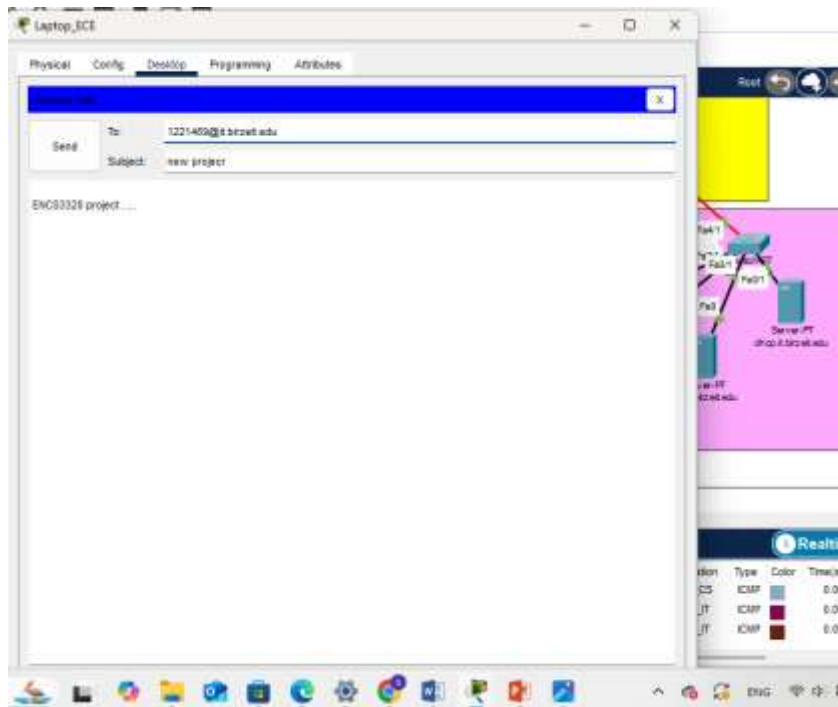


Figure 51 : write email to it.birzeit.edu

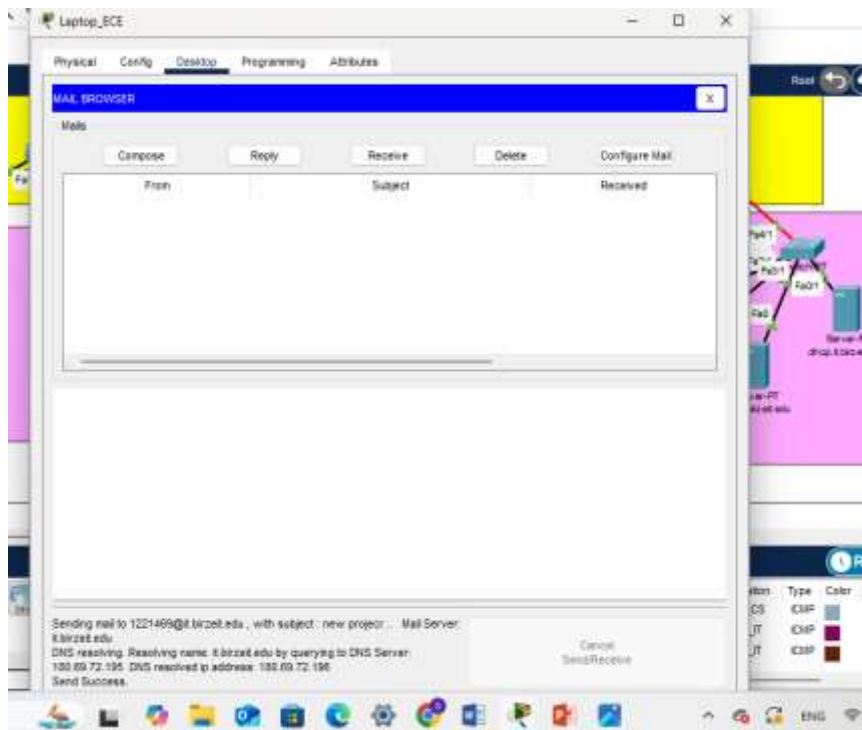


Figure 52 : Figure 47 : send it.birzeit.edu to another one

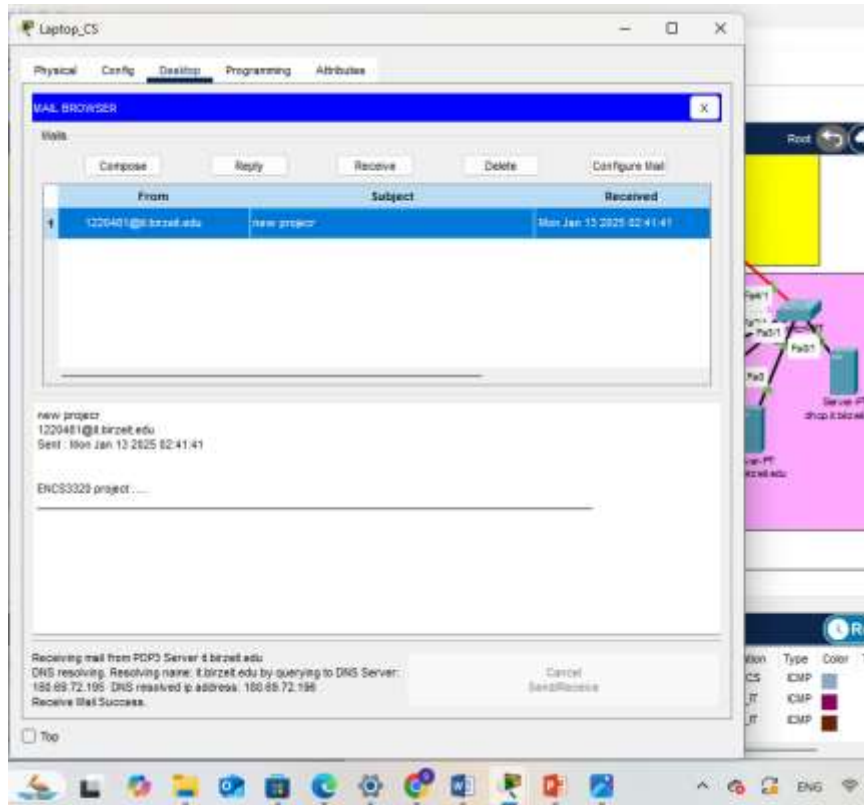


Figure 53 : receive it.birzeit.edu from another one

3. Home Network :

3.1 Static IP configuration for PC_Home.

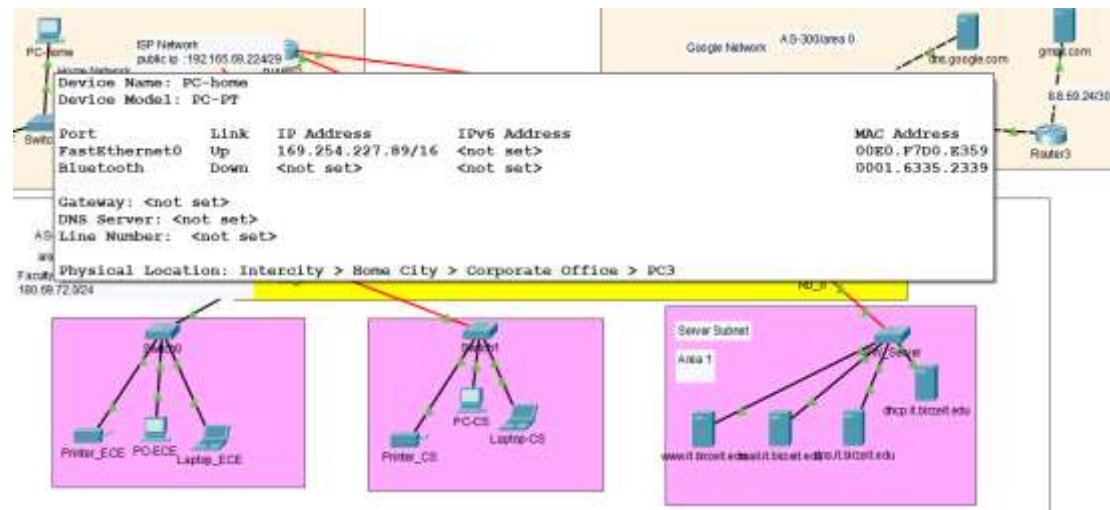


Figure 54 :IP for PC_HOME

3.2 Dynamic IP configuration for a home network end device.

Dynamic IP to laptop_home → 192.168.69.3

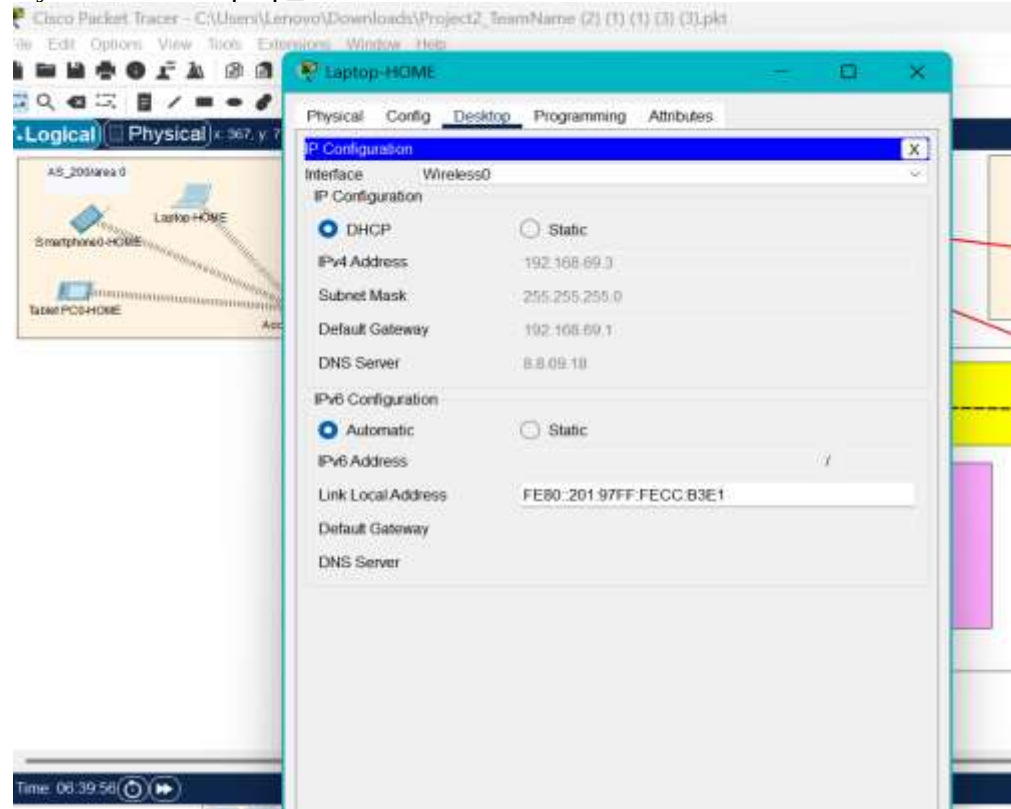


Figure 55 :IP for Laptop_home

Dynamic IP to smartphone_home → 192.168.69.2

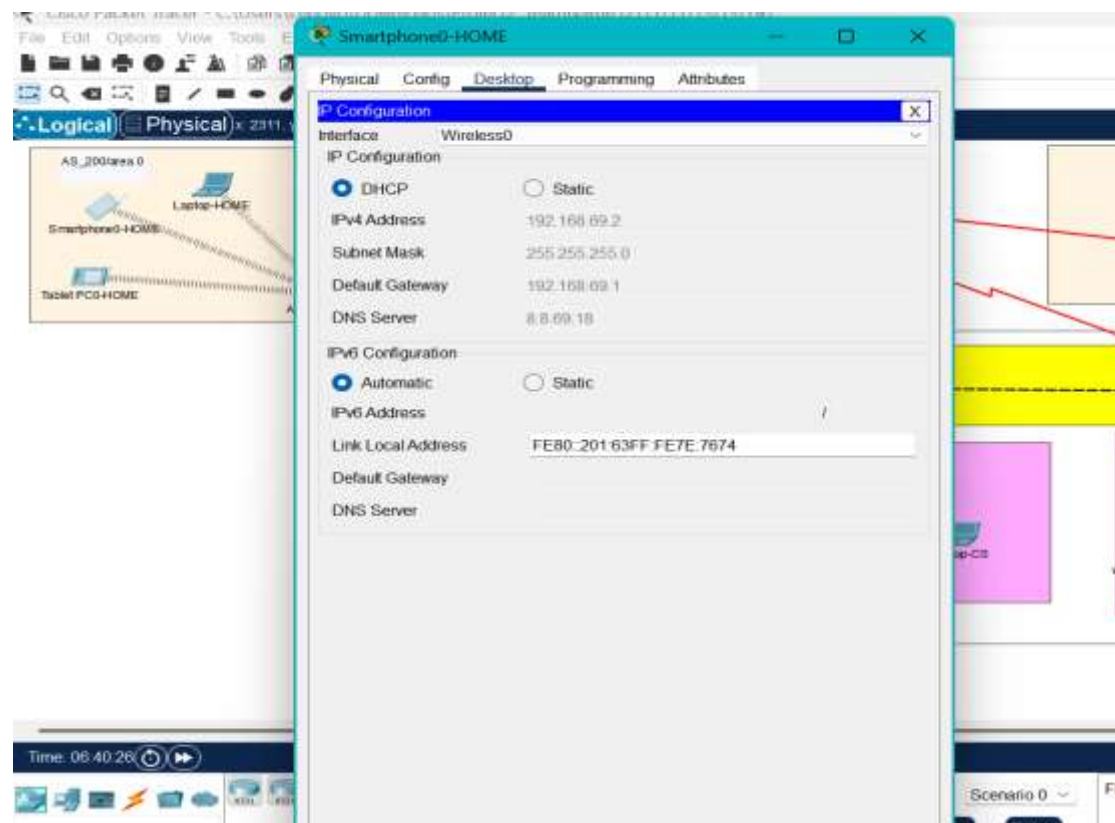


Figure 56:IP for smartphone_home

Dynamic IP to Tablet_home → 192.168.69.4

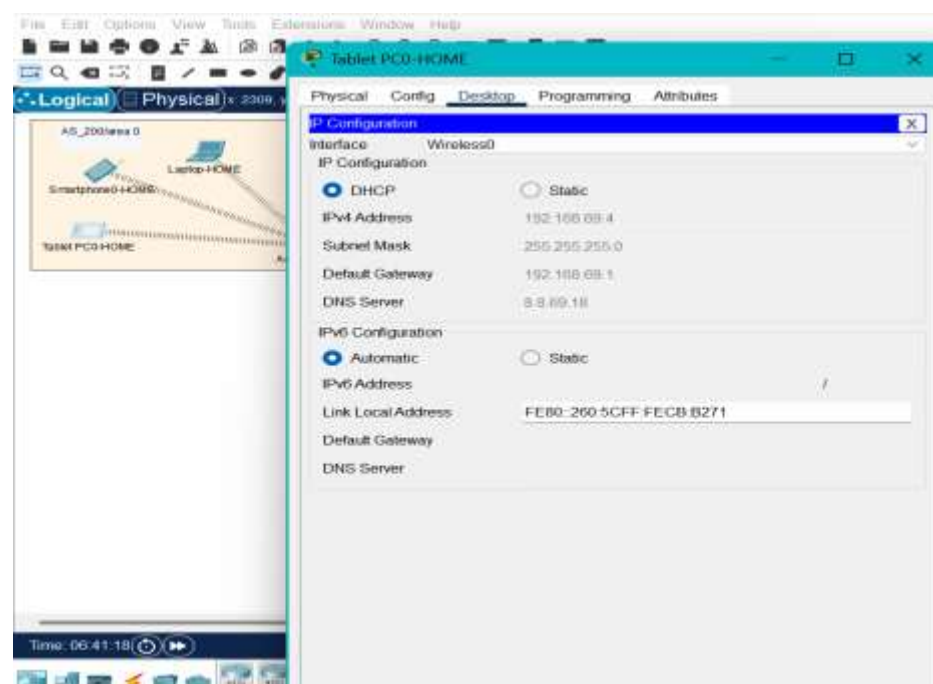


Figure 57 : IP for tablet_home

Dynamic IP to PC_home → 192.168.69.5

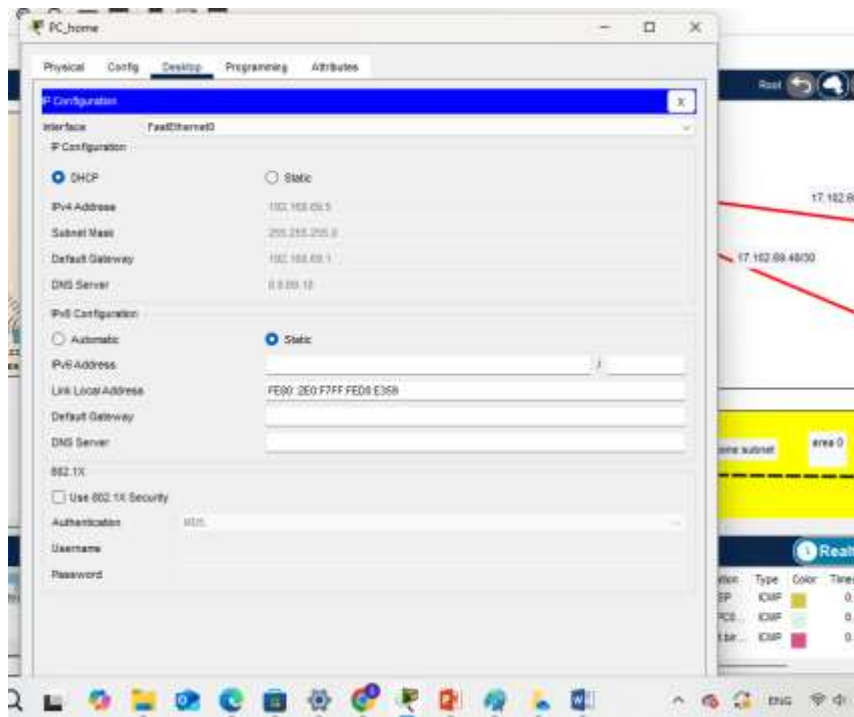


Figure 58: Dynamic IP for PC_Home

3.3 : Static IP configuration of the R_Home router port.

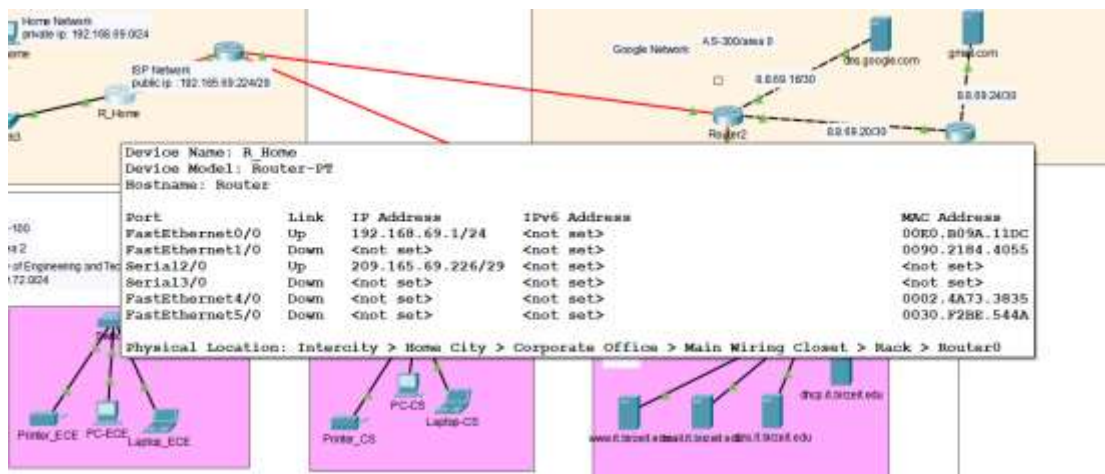


Figure 59 : IP OF R_home

3.4 NAT table of R_Home router from the command:

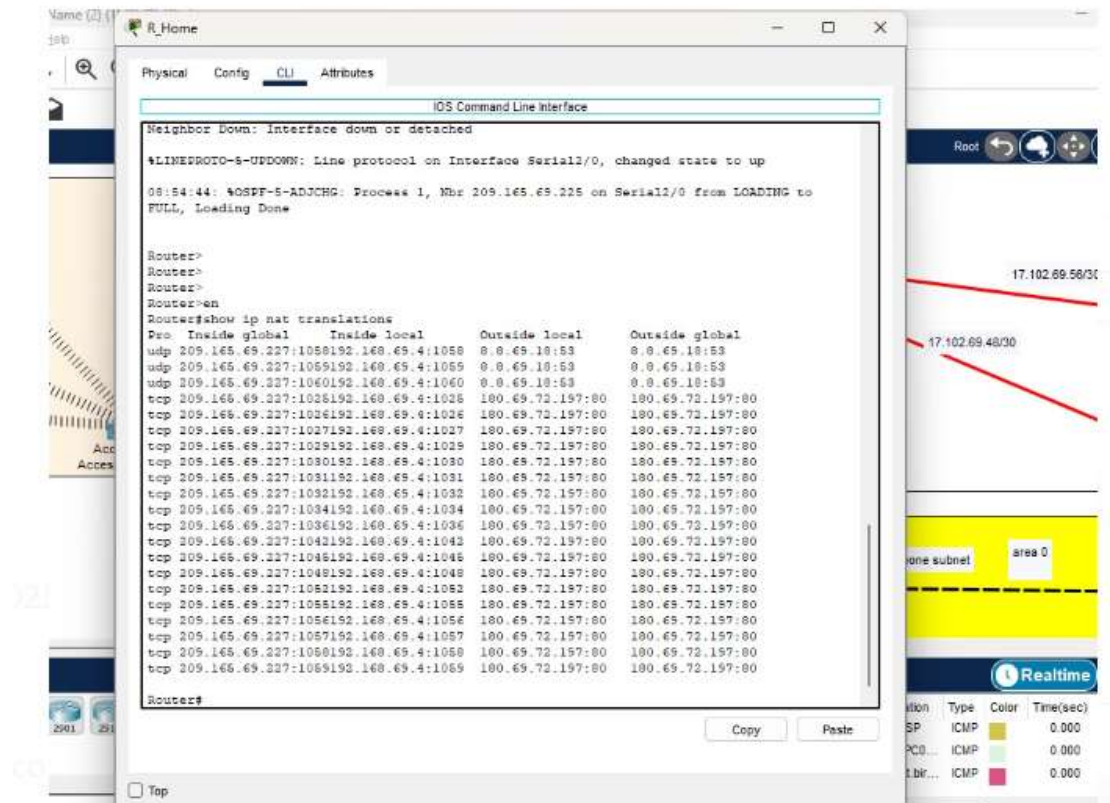


Figure 60:NAT table of R_Home

3.5 Successful Ping results between home network devices.

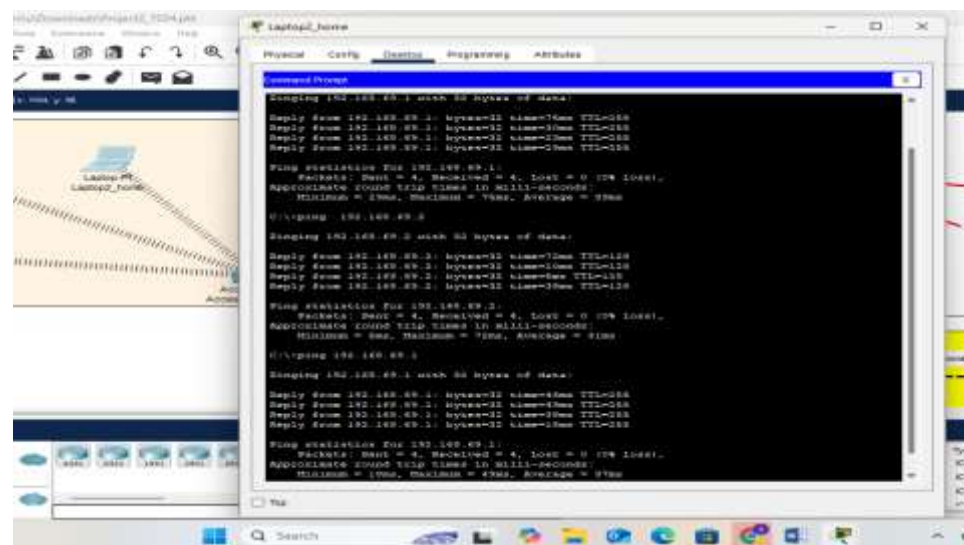


Figure 61: success ping 1

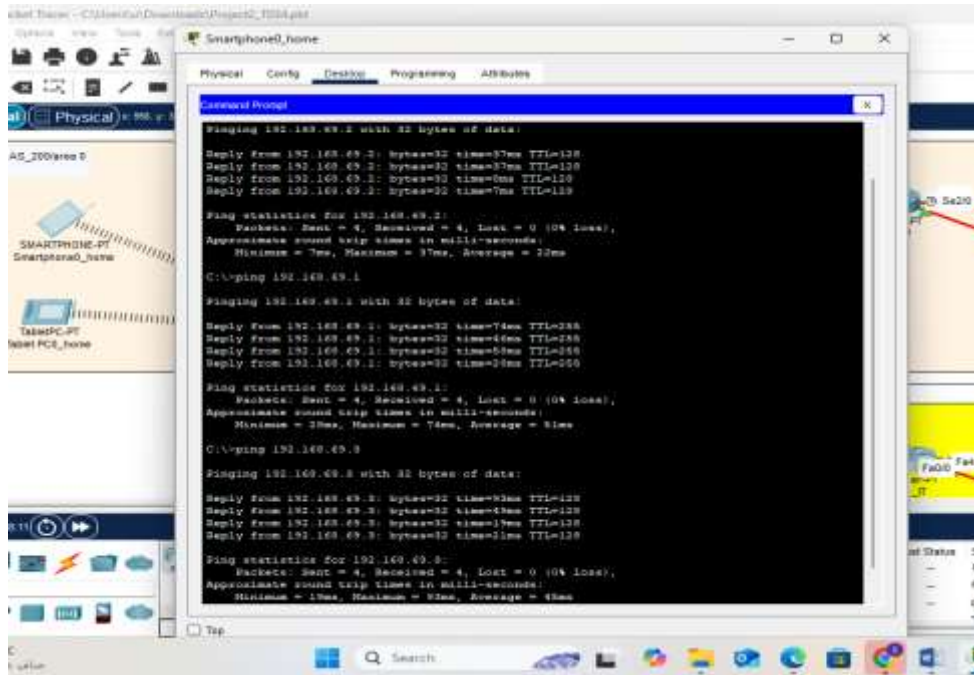


Figure 62:success ping 2

From the above pictures ,we have a successful Ping results between home network devices. And There is no packet loss

3.6 Wireless network configuration for ENCS3320Home.

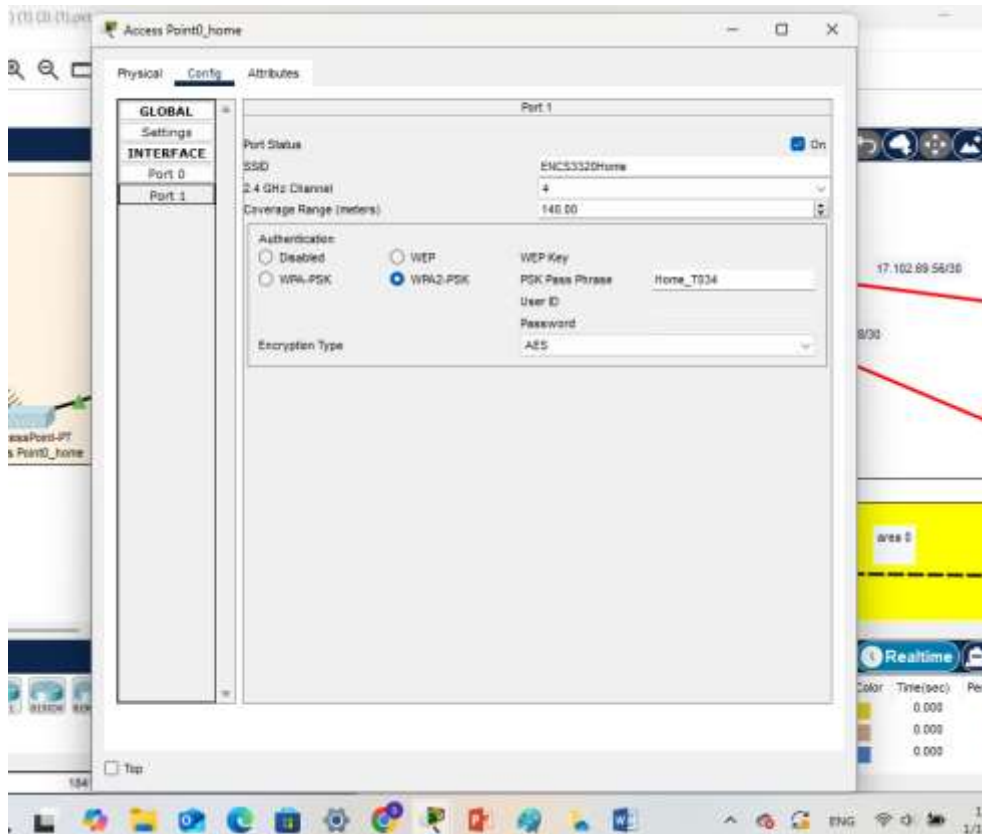


Figure 63 :wireless Network

3.7 Successful access to www.it.birzeit.edu from a home network end device :



Figure 64 :Successful access to www.it.birzeit.edu from laptop_home

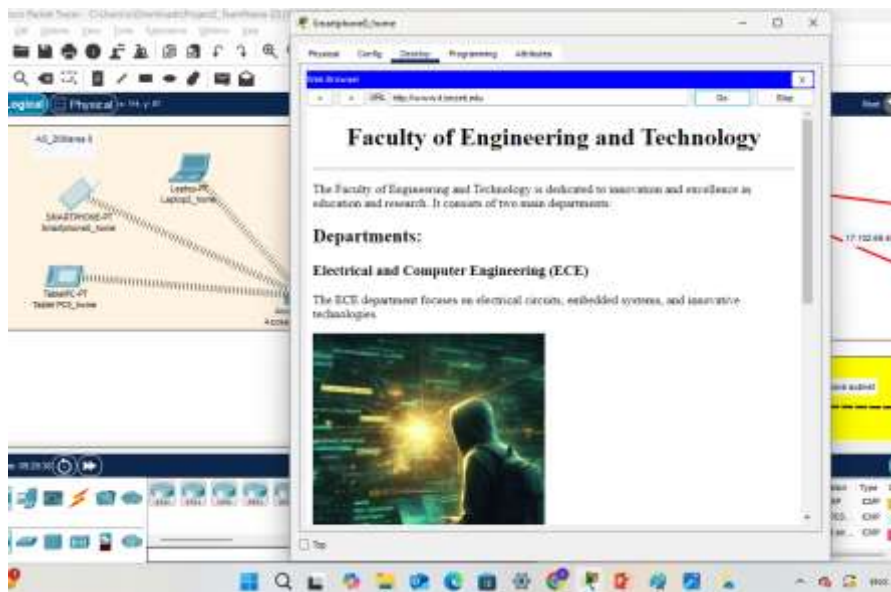


Figure 65:Successful access to www.it.birzeit.edu from samrtphonnr_home

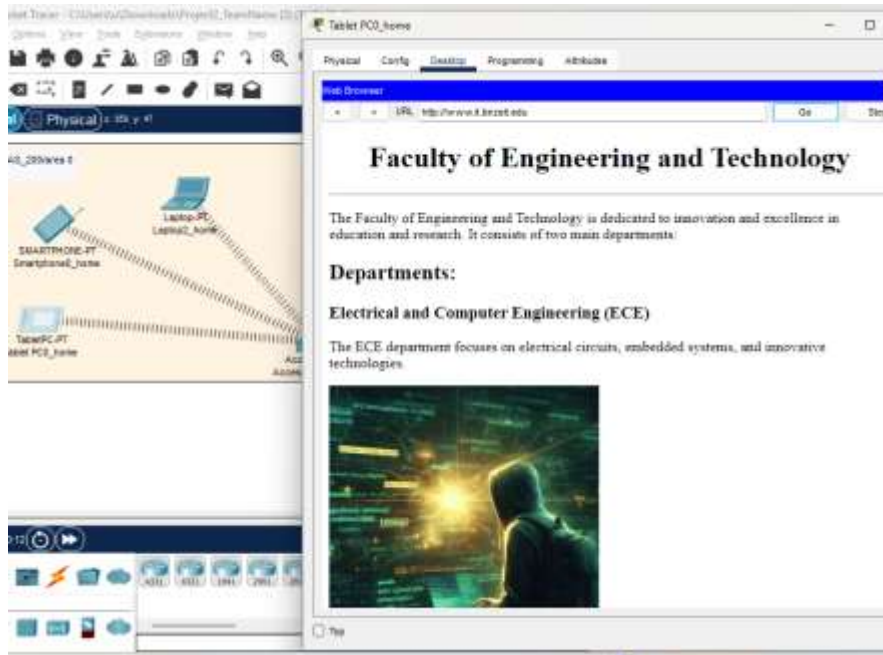


Figure 66 :Successful access to www.it.birzeit.edu from tablet_home



Figure 67 :Successful access to www.it.birzeit.edu from PC_home

3.8 : Email client configuration for gmail.com and it.birzeit.edu accounts.

The gmail client in laptop_home & PC_home :

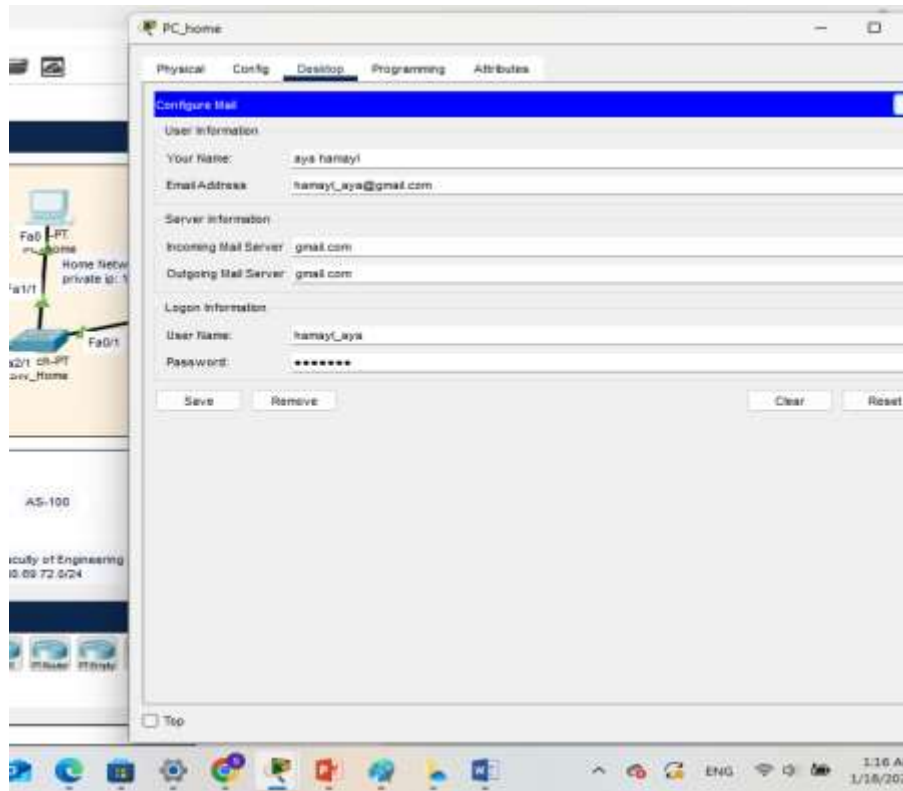


Figure 68: gmail client in PC_home

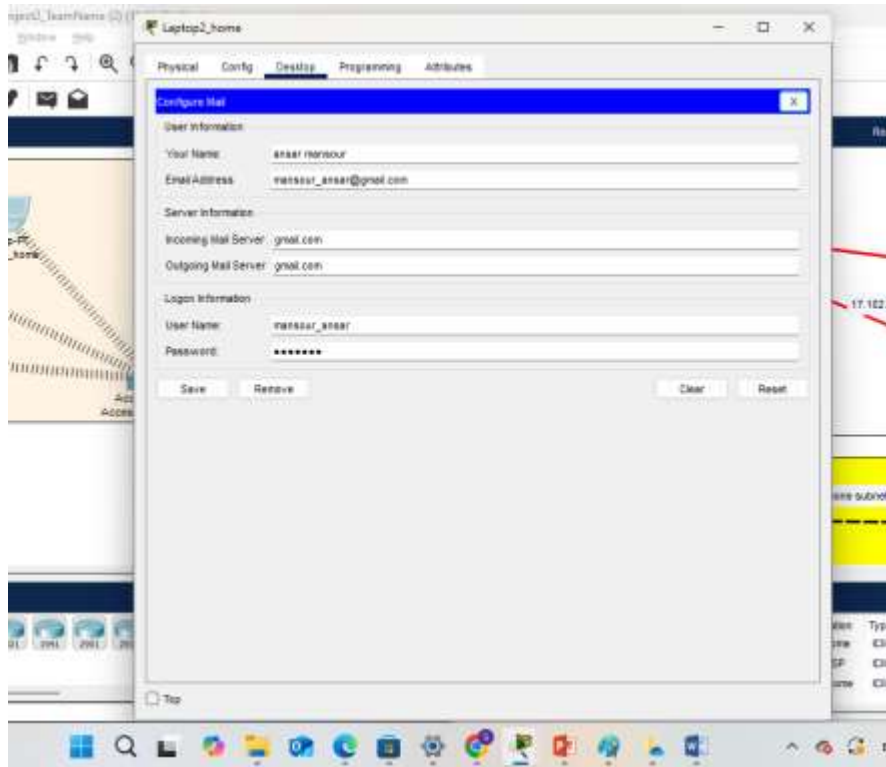


Figure 69 :gmail client in laptop_home

The it.birzeit.edu client in tablet_home & smartphone_home :

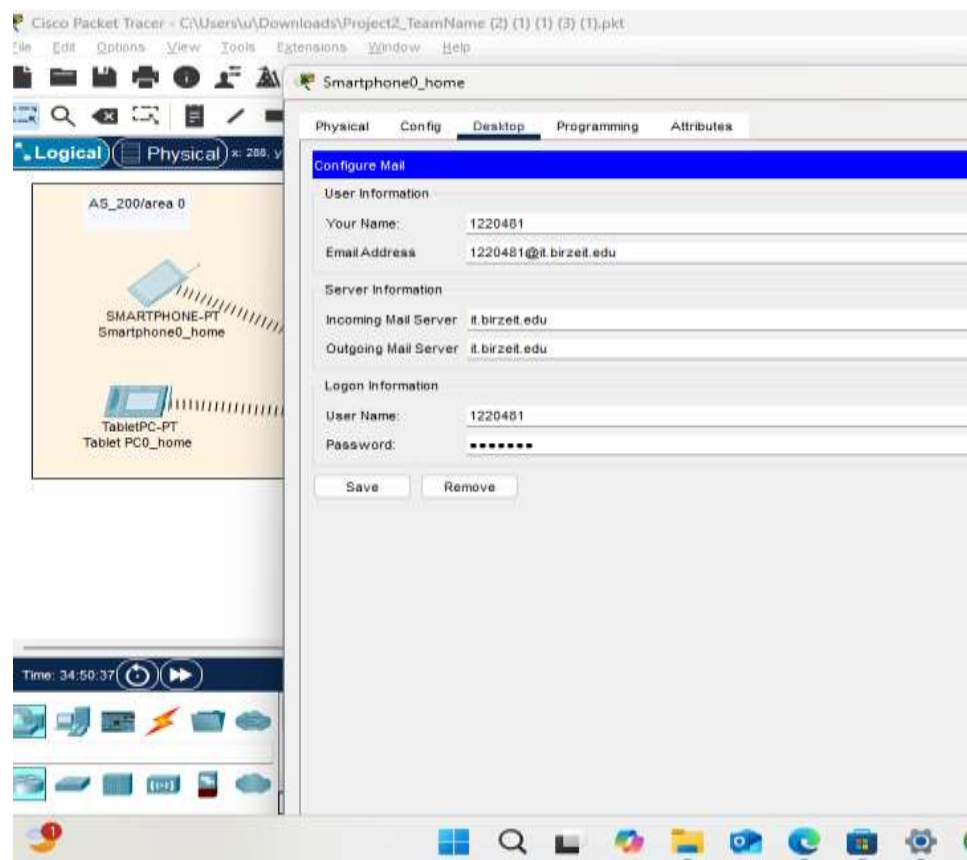


Figure 70 :it.birzeit.edu client in smartphone_home

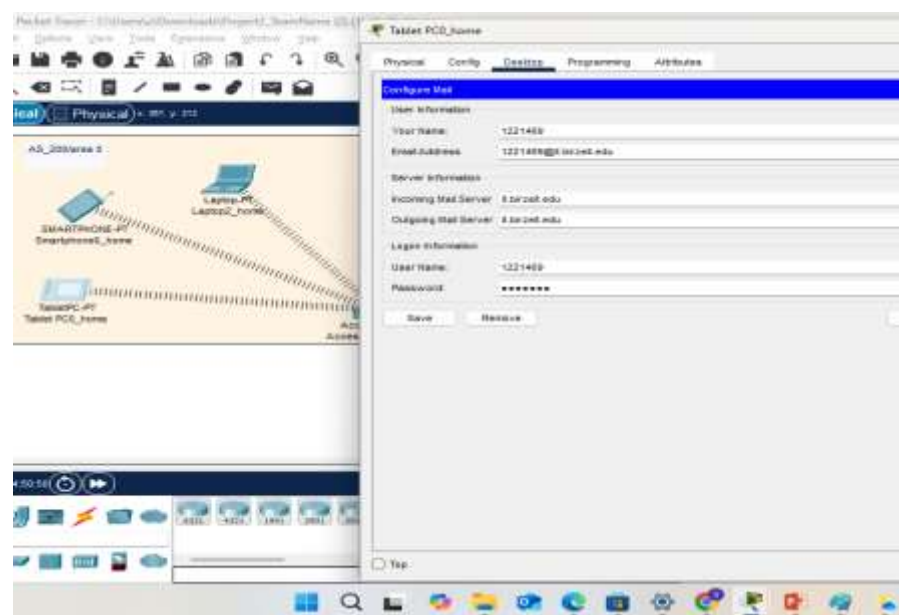


Figure 71 : it.birzeit.edu client in tablet_home

3.9 : Successful sending and receiving of emails:

3.9.1 : Between gmail.com accounts within the Home-ISP network.

First we sent email from email Desktop by Laptop_home who own Mansour_ansar@gmail.com to PC_home hamayl_aya@gmail.com

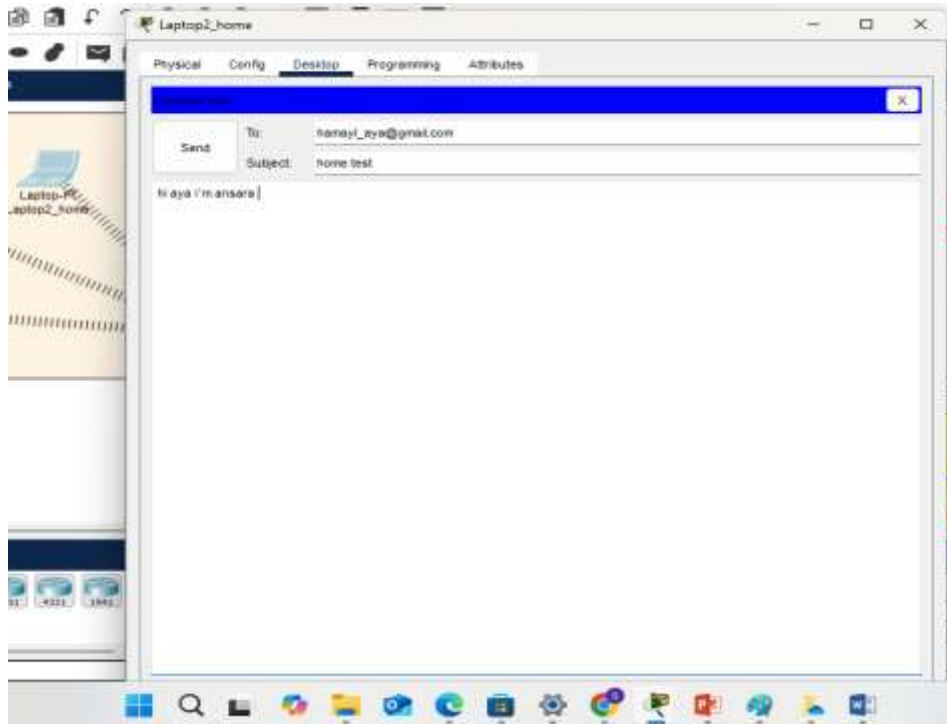


Figure 72: write email to send

The email was send success

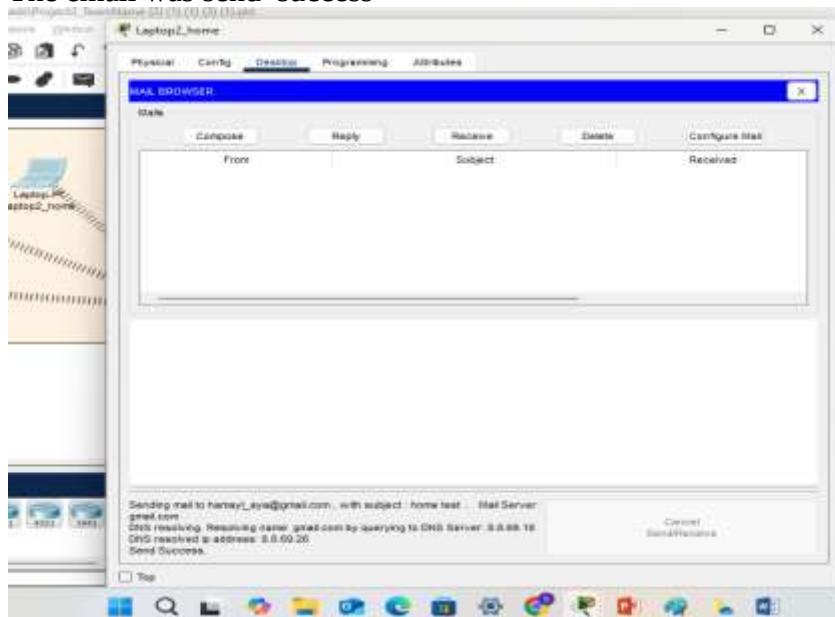


Figure 73 :email was success send

The receiving side received the email successfully:

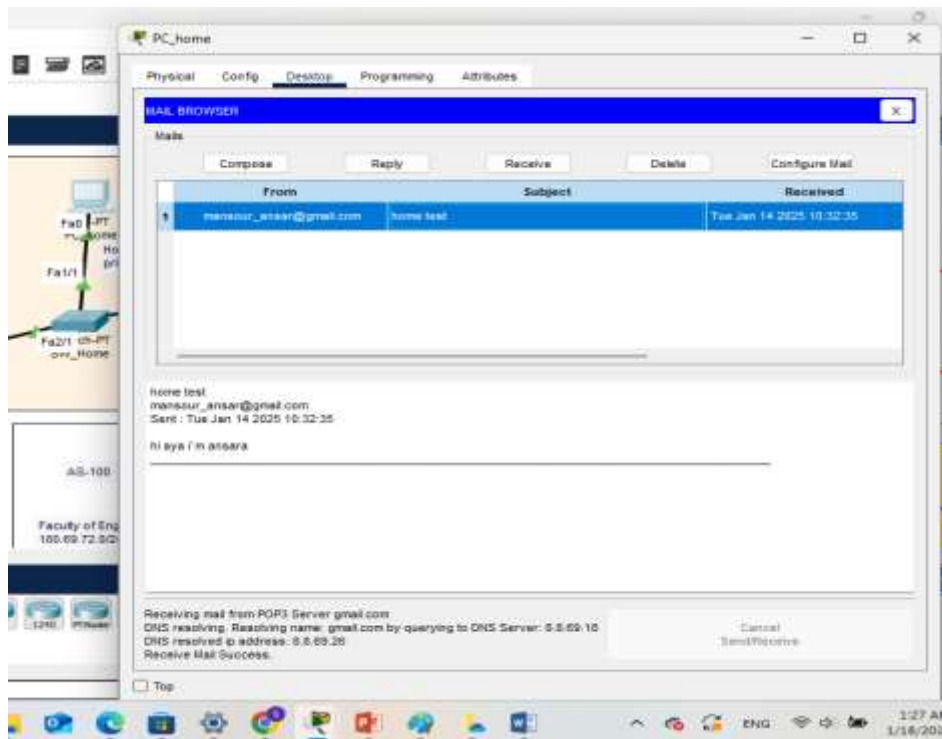


Figure 74 ;receiving email

3.9.2 : Between it.birzeit.edu accounts within the Home-ISP network.

First we sent email from email Desktop by smartphone_home who own 1220481@it.birzeit.edu to tablet_home 1221469@it.birzeit.edu

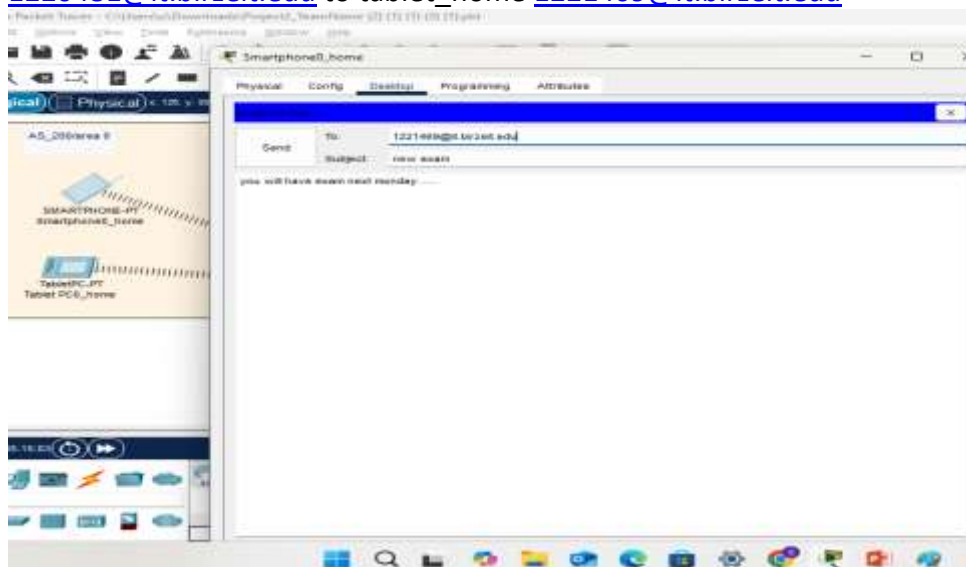


Figure 75:writer it email

The email was success to send

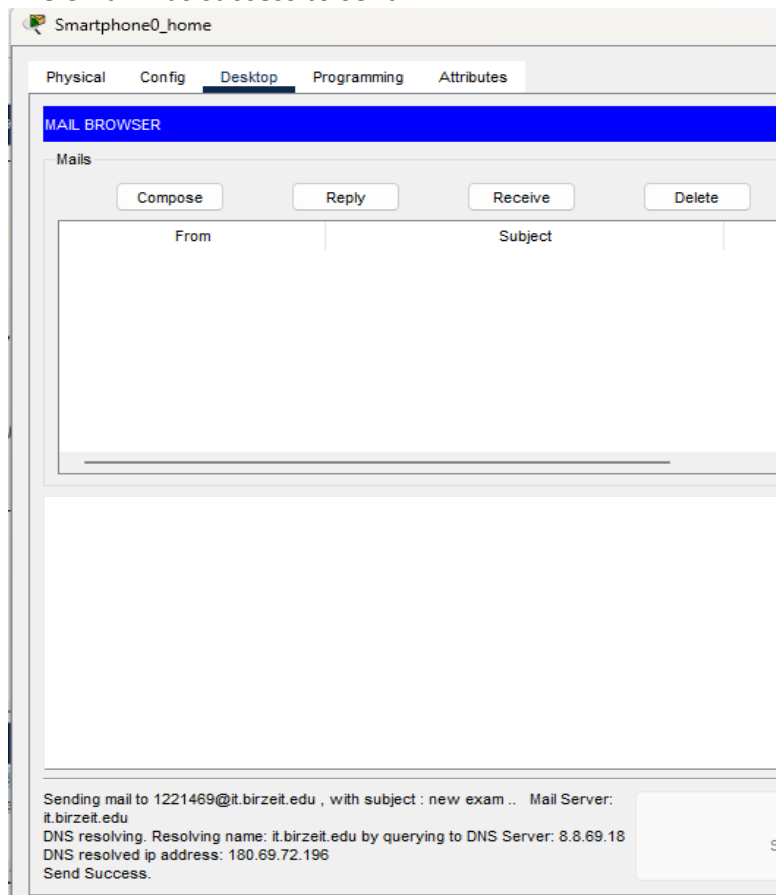


Figure 76:email was success to send

The receiving side received the email successfully:

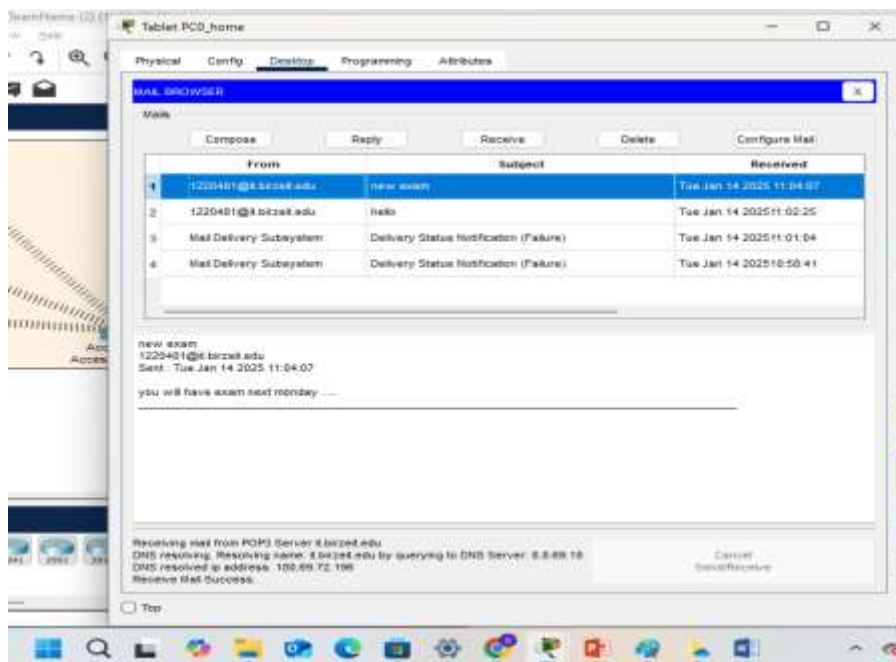


Figure 77:receiving email

3.9.3 : Between gmail.com and it.birzeit.edu accounts within the Home-ISP network.

We send the email from gmail.com client in laptop_home (mansour_ansar@gmail.com) to it.birzeit.edu client in tablet_home (1221469@it.birzeit.edu) so we write a message and it receive success

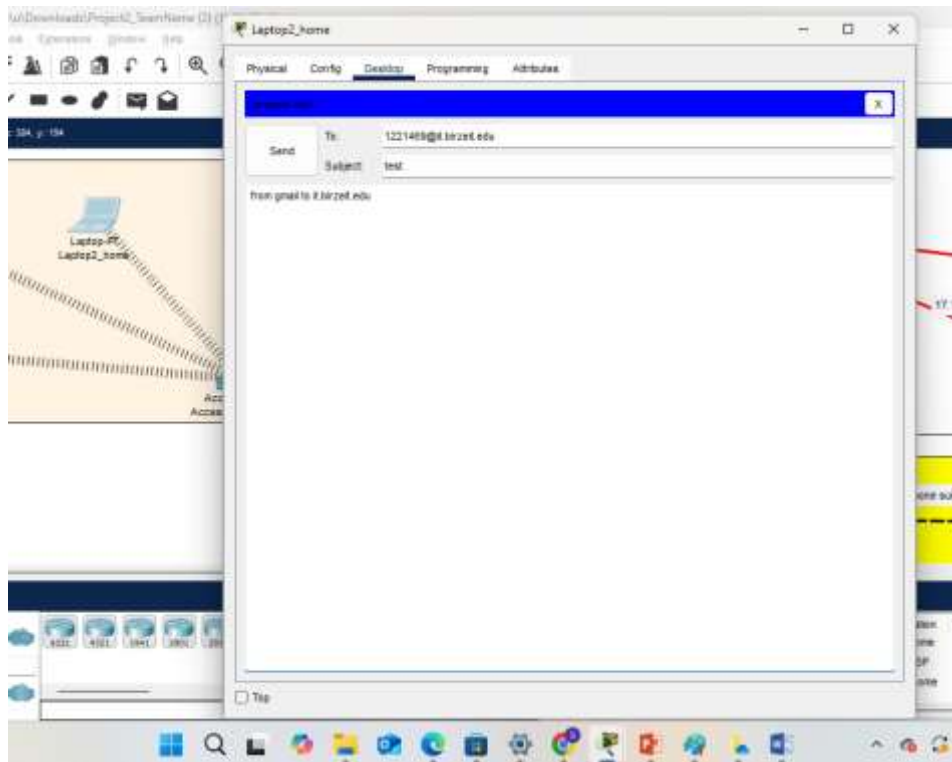


Figure 78 :write email

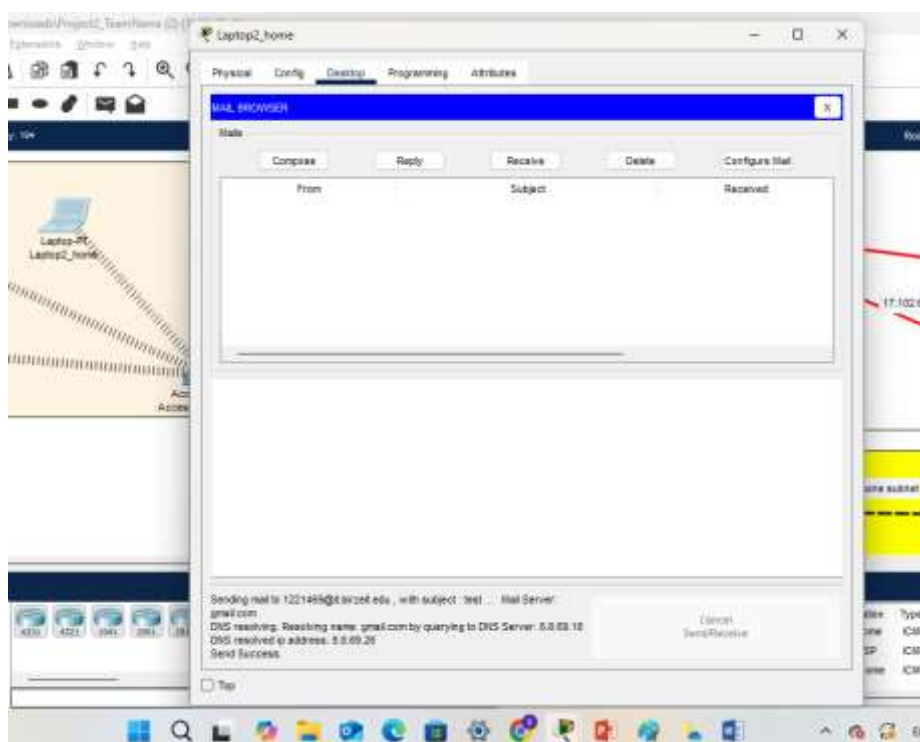


Figure 79:success email sand

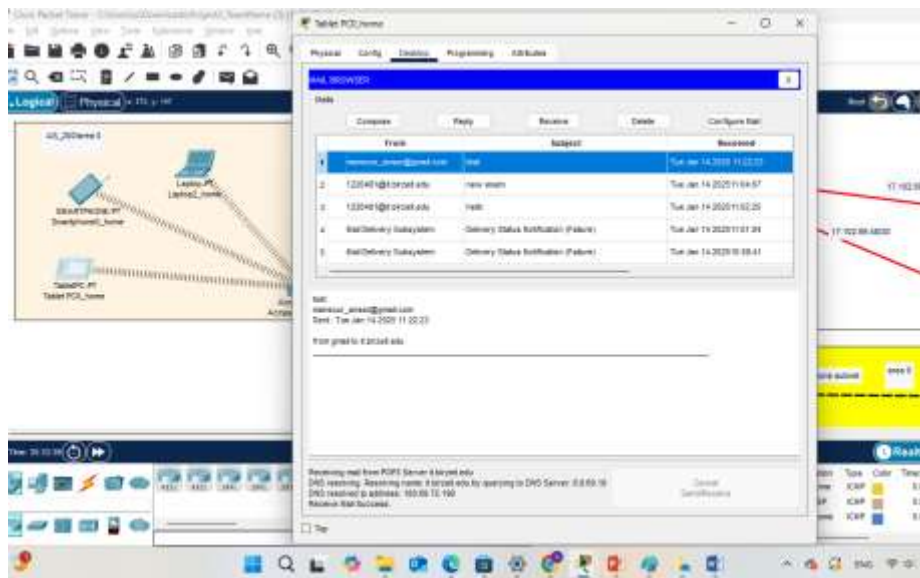


Figure 80:th receive email

We send the email from it.birzeit.edu client in smartphone_home (1220481@it.birzeit.edu) to gmail.com in PC_home (hamayl_aya@gmail.com) so we write a message and it receive success

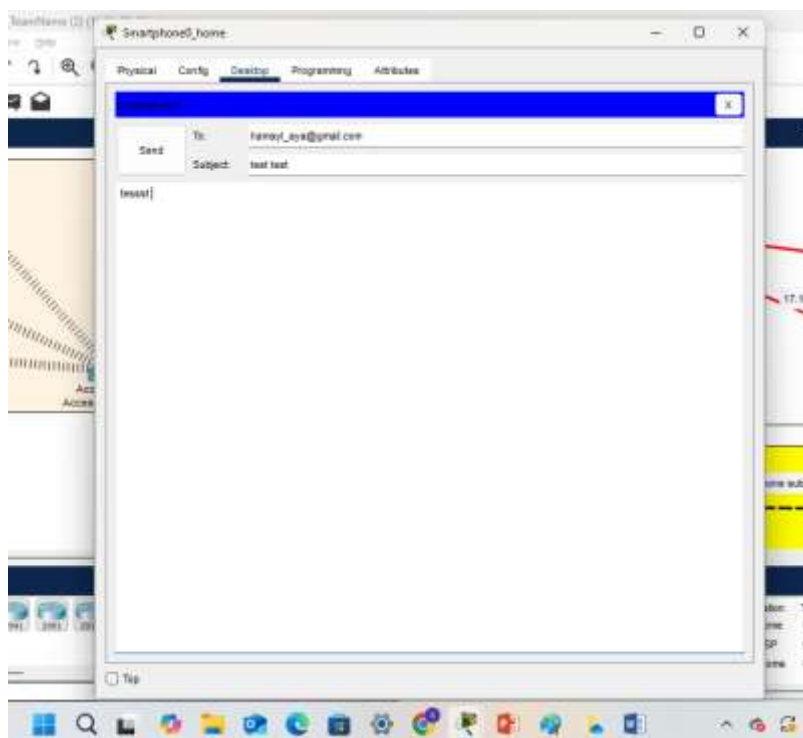


Figure 81:send email

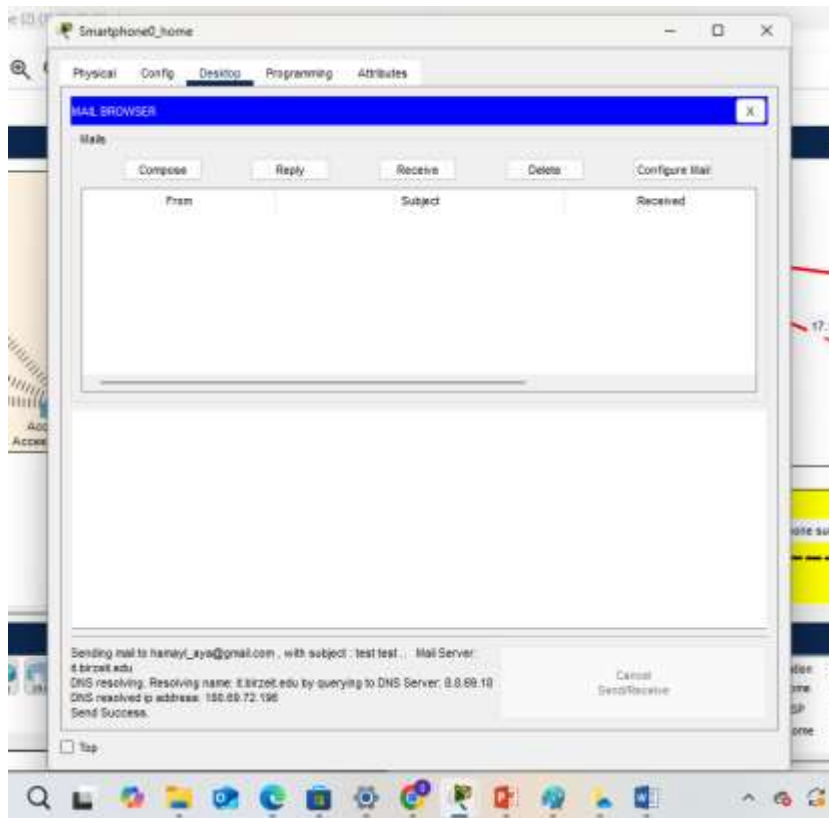


Figure 82:success send

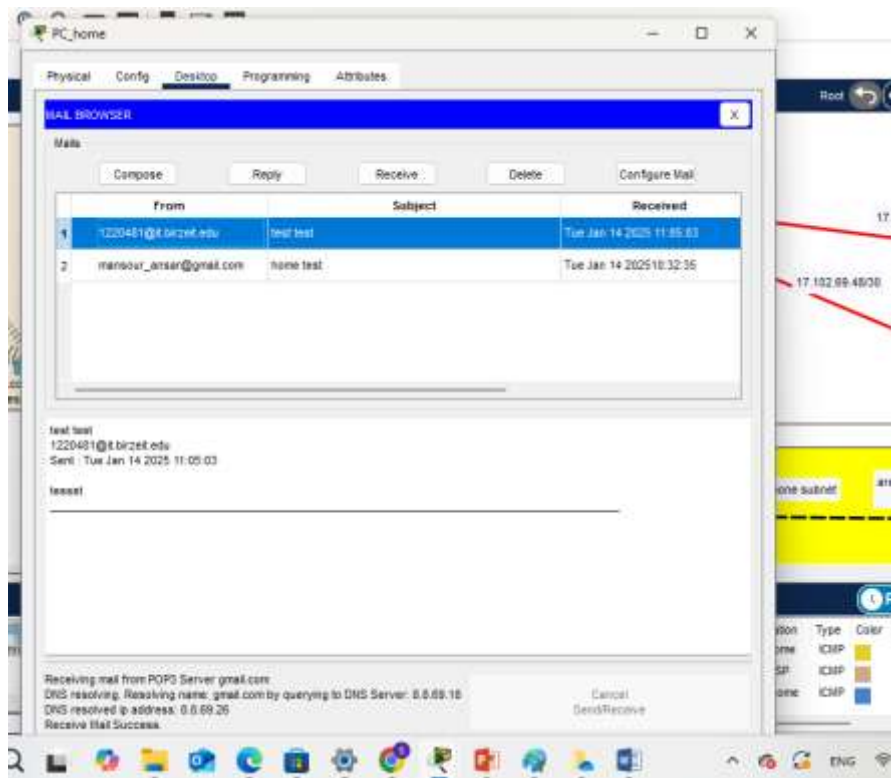


Figure 83 :success receive

3.9.4 : Between Gmail accounts within the Home-ISP network and the Faculty of Engineering and Technology network.

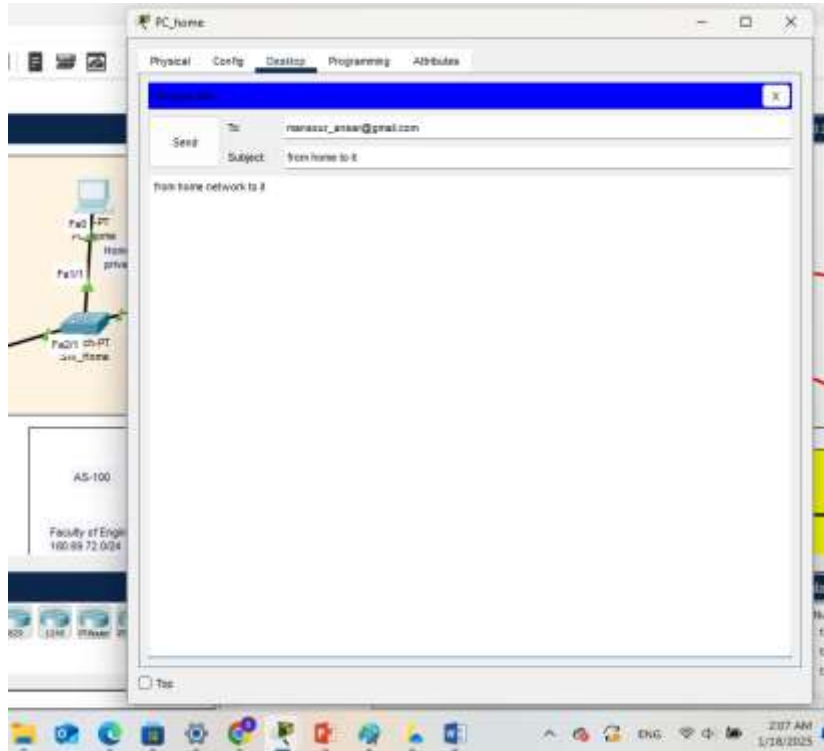


Figure 84 :write email

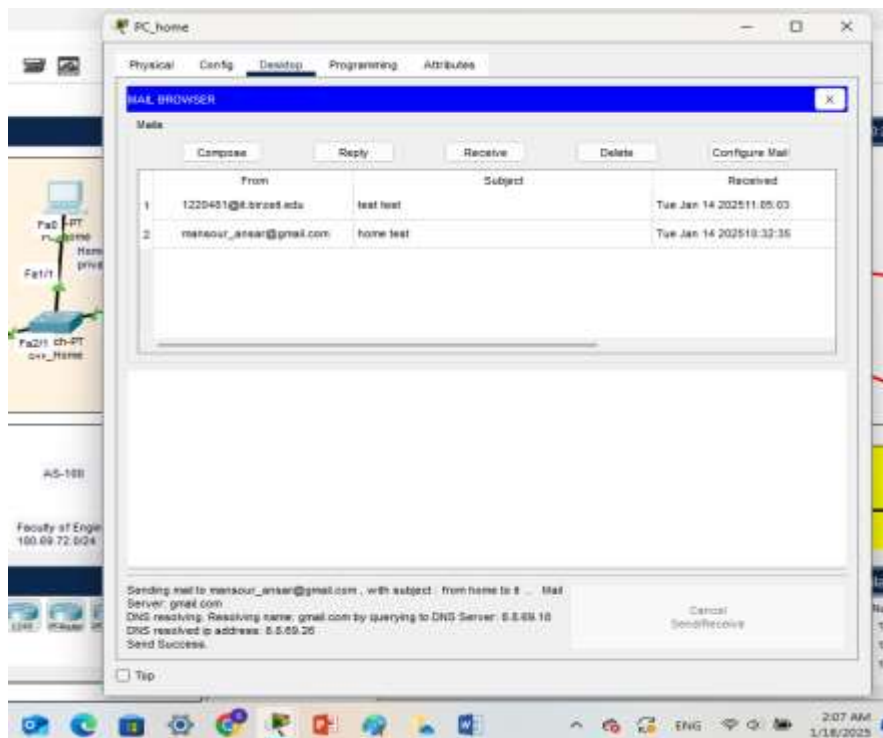


Figure 85 :success send

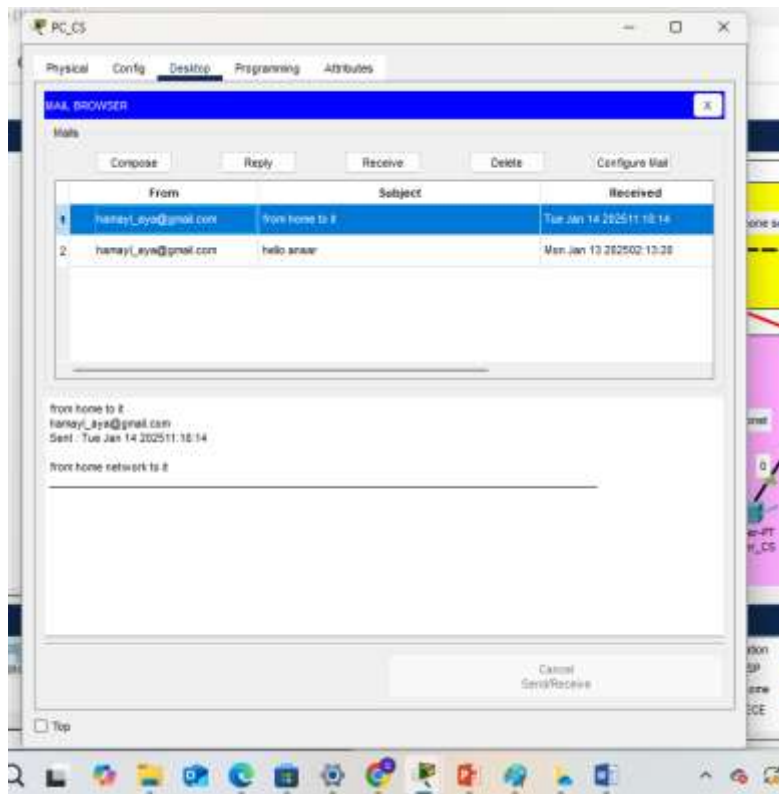


Figure 86 :success receive

3.9.5 Between Birzeit accounts within the Home-ISP network and the Faculty of Engineering and Technology network

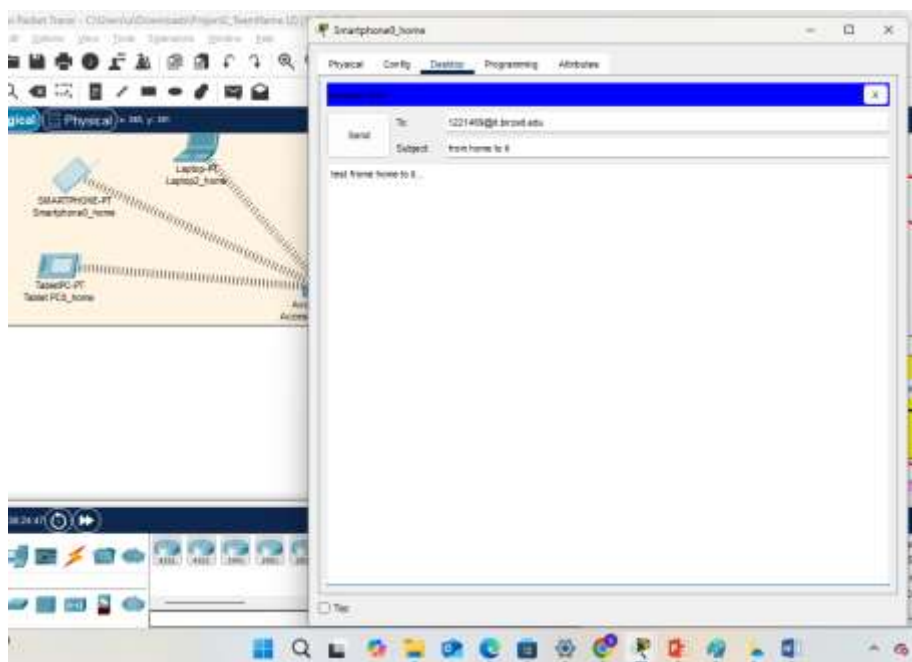


Figure 87 :write email

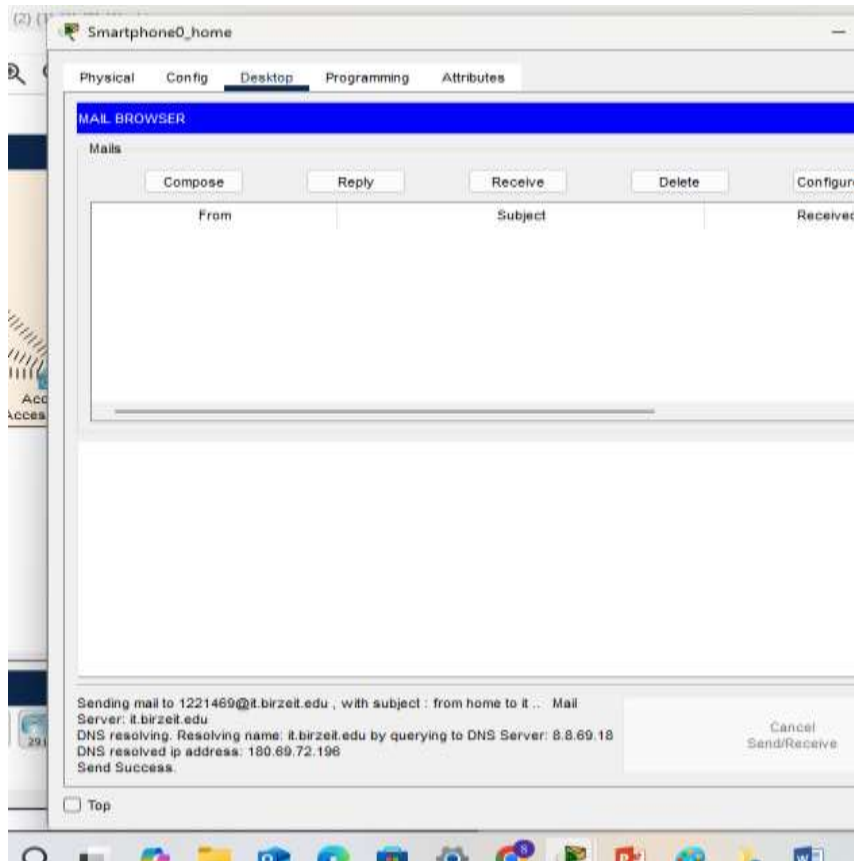


Figure 88 :success send

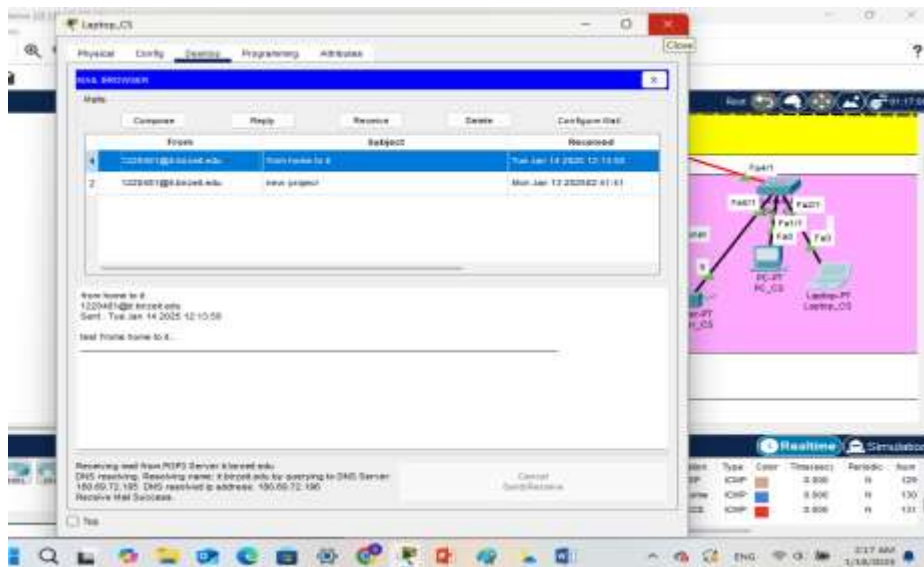
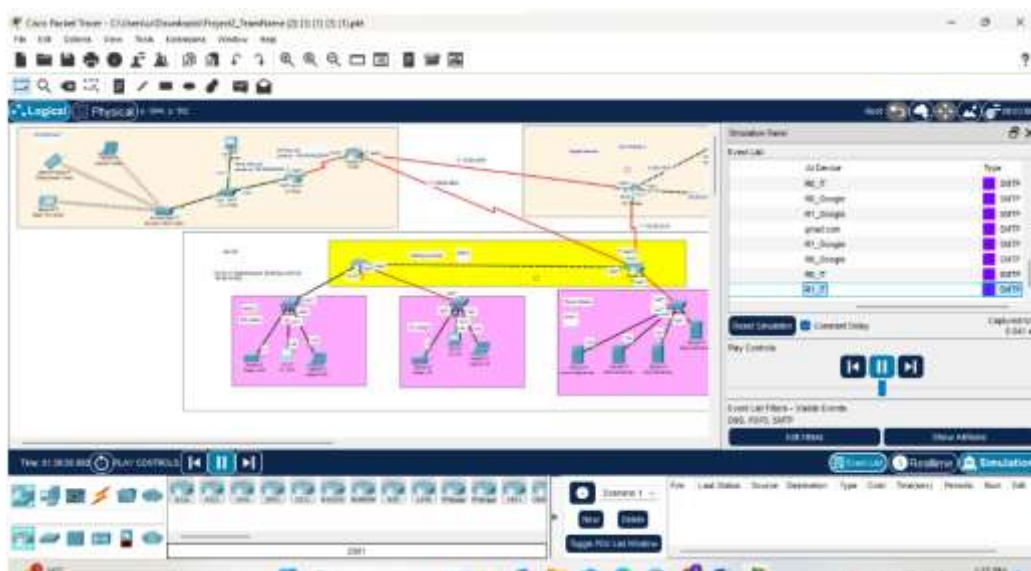
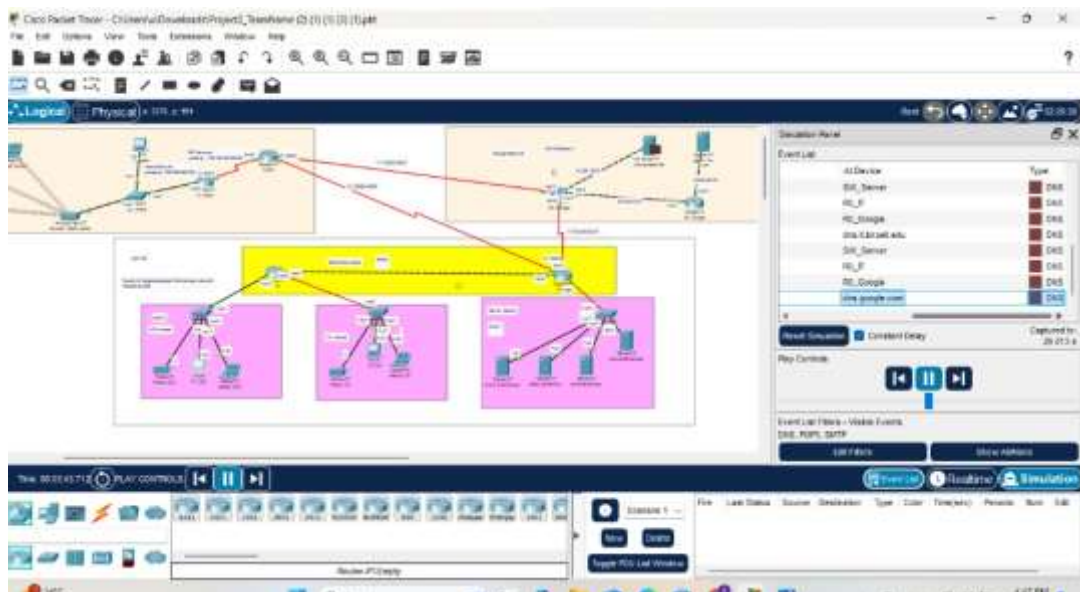


Figure 89:succes receive

SIMULATION➔



1. DNS Query Messages:

- Initially, DNS messages were observed, indicating that the source device queried for the domain names (gmail.com and it.birzeit.edu) to resolve their corresponding IP addresses.
- The event log in the simulation panel shows DNS requests and responses, confirming successful name resolution.

2. SMTP Messages:

- After DNS resolution, SMTP (Simple Mail Transfer Protocol) messages were exchanged between the devices. These messages represent the process of sending the email from the Gmail account to the IT Birzeit email server.
- The event log lists several SMTP messages, with the source and destination addresses indicating the mail servers involved in transferring the email.

3. POP3 Messages:

- Following the SMTP messages, POP3 (Post Office Protocol) messages were used to retrieve the email from the destination server to the email clients.
- Two email clients were configured: one in the Home-ISP network and the other in the Faculty of Engineering and Technology network. The simulation log confirms the flow of POP3 messages to both clients.

Findings and Observations:

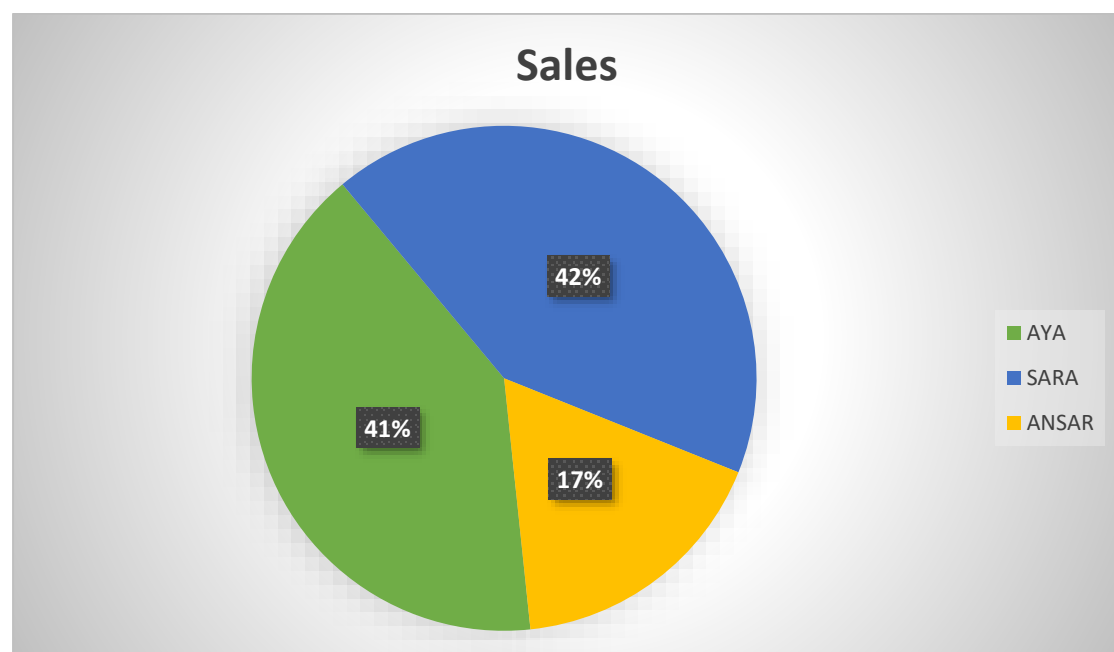
1. Email Retrieval:

- The email was successfully retrieved on both devices.
- This indicates that the DNS resolution, SMTP email transfer, and POP3 email retrieval processes were correctly configured and functional.

Teamwork :

Ansar Mansour worked on setting up a web server and configuring DHCP in the Home network in collaboration with Sara Shrouf. Additionally, Ansar was responsible for tasks such as IP addressing and subnetting in the Faculty of Engineering network, configuring DHCP in the same network, However, due to multiple errors in his work, Aya and Sara reviewed the configurations, decided to discard Ansar's work entirely, and restarted the implementation from scratch. They successfully reconfigured and completed all tasks, ensuring that the network setup was fully operational and error-free.

AYA Hamayl and SARA shrouf : both completing the remaining tasks and Completion of all project requirements successfully. and dividing the report between them.



References :

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- [2] <https://www.techtarget.com/searchnetworking/answer/Static-and-dynamic-routing#:~:text=Static%20routing%20uses%20a%20single,available%20routes%20to%20the%20destination.>
- [3] <https://www.geeksforgeeks.org/open-shortest-path-first-ospf-protocol-states/>
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- [10]